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Master's Thesis

Predicting Crowdfunding Campaign Success: A Multimodal Machine Learning & Deep Learning Approach

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Abstract

Crowdfunding has become an important alternative financing mechanism in recent decades, but a large fraction of campaigns still fail, highlighting the need for reliable methods to forecast the outcome of these campaigns. This thesis aims to explore the use of multimodal data and machine learning techniques to predict the success of Kickstarter campaigns based only on pre-launch information. Using a dataset of 40,301 Kickstarter campaigns, this study evaluates a wide range of approaches, including traditional machine learning and deep learning architectures, multimodal fusion strategies and a large language model. It provides a comprehensive framework for the integration of structured metadata, textual and visual features and systematically compares the results across multiple modeling approaches and configurations. Furthermore, it provides a new insight into the effect of different text processing approaches on predictive performance by introducing a novel comparison of structured and unstructured text representations. It extends the use of LLMs to structured prediction tasks and evaluates not only their ability to predict the outcome of campaigns but also their ability to generate human-readable explanations. Furthermore, this study combines predictive modeling with interpretability methods like SHAP analysis and LLM-based reasoning to enhance the interpretability and practical relevance of the results.

The results indicate that strong predictive performance can be achieved using structured metadata only with models such as Logistic Regression, SVM and XGBoost. The inclusion of textual features further improves their performance, especially when the TF-IDF representations are combined with the metadata. This configuration consistently outperforms either of the models employing only metadata and more complex deep learning models. Logistic Regression with TF-IDF and metadata gave the best overall results (ROC-AUC=0.904). This shows that the majority of the predictive signal can be extracted in a high-dimensional linearly separable feature space. On the other hand, multimodal models that integrated visual features only achieved marginal improvements that indicate the limited additional value of images in this context and form. The feature importance analysis performed using SHAP values and model coefficients highlights key features of campaign design, including the goal amount, reward structure, campaign duration and the central role of media assets. While the LLM does not achieve the accuracy of traditional models, it provides human-interpretable explanations that are consistent with the quantitative results, demonstrating the value of the LLMs complementary interpretability tools.

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The code developed and used in this study is available on GitHub:
https://github.com/bzsofi00/CrowdfundingSuccessPrediction.git

The datasets used in this study can be found on: Datasets

1. Introduction

Crowdfunding has become and continues to be an important financing mechanism for entrepreneurs, creative professionals, and new businesses. It is an alternative form of financing and refers to the practice of raising funds from a large number of individuals, the crowd (Belleflamme et al., 2014). It usually takes place on various websites without involving official financial institutions (Mollick, 2014). It has proliferated with the appearance of Web 2.0 and the rise of digital platforms such as Kickstarter (Kleemann et al., 2008). These platforms facilitate direct interactions between project creators and potential backers, allowing creators to showcase their ideas, share updates, and engage with their audience during the campaign. As of August 2025, Kickstarter has hosted more than 651,000 projects and raised $8.99 billion with a success rate of 41.98% (Statista, 2025b). Such a high failure rate indicates that crowdfunding is still very uncertain, and there is an increasing need for reliable ways to predict the campaign's outcome and get a better understanding of the factors that affect it. Existing research has made important contributions to finding predictors of crowdfunding success, especially by using structured campaign metadata and textual descriptions. They have identified that features like funding goal, the duration of the campaign, reward structures and linguistic characteristics of campaign descriptions can influence funding outcomes. Recently, machine learning and deep learning models have been applied to this problem, achieving increasingly stronger predictive performance. However, several important limitations still remain. First of all, many studies focus on structured data or textual information, while visual content, such as images, remains underexplored, although it is a central element of campaign presentation. Moreover, while much attention is given to prediction accuracy, the aspects of explainability and interpretability are often overlooked or regarded as of lesser importance. This oversight reduces the usefulness of these models for practical decision-making. Many approaches rely on post-launch dynamics such as the early funding process, which limits their usefulness in real-world situations where predictions are most valuable before a campaign is launched. Furthermore, while recent approaches started to explore generative models, their role in structured prediction tasks and their ability to provide meaningful reasoning remain highly underexplored.

In response to these persistent challenges, this study provides a comprehensive analysis of predicting crowdfunding success by combining structured metadata, textual characteristics, and visual elements within a single modeling framework. It evaluates the predictive performance of different models (classical machine learning, deep learning and multimodal) and includes explainability methods to provide practical insights on the main drivers of campaign success. It also investigates the predictive

power of large language models (LLMs) and their ability to provide coherent qualitative reasoning to justify their predictions. The analysis is limited to pre-launch information, making the results more practically relevant for campaign creators and platform designers.

Based on these objectives, the study is directed by the following main research question:

How can multimodal data and machine learning techniques be used to effectively predict the success of crowdfunding campaigns using only pre-launch information?

To answer this overarching question, the following sub-questions are examined:

- How well can crowdfunding success be predicted using only structured metadata, and which models perform best in this setting?
- How does incorporating textual features improve predictive performance compared to metadata-only models?
- How do multimodal approaches, including visual features, provide additional predictive value beyond textual and metadata-based models in crowdfunding success prediction?
- Which features are the most important in predicting crowdfunding success, and how can explainability methods improve the understanding of model predictions?
- How effectively can large language models predict and explain crowdfunding campaign success compared to traditional machine learning and deep learning approaches using text, image, and metadata features?

The current study makes several contributions to the literature. It first provides a thorough multimodal analysis by combining structured, textual and visual data and then studies different fusion strategies to evaluate the effectiveness of combining such data types. The paper provides a systematic comparison between traditional machine learning and deep learning approaches and introduces a novel comparison between structured and unstructured textual representations. It also includes and extends the application of large language models in crowdfunding prediction, showing that these models are able to not only perform classification tasks but also generate human-readable explanations, thereby bridging the gap between prediction and interpretation. Furthermore, it combines predictive modeling with interpretability methods (e.g., SHAP and LLM-based reasoning), providing both robust quantitative performance and relevant qualitative insights. It also emphasizes prediction before launch, which enhances the practical relevance of the results for real-world decision-making.

2. Theoretical Background

2.1. Crowdfunding

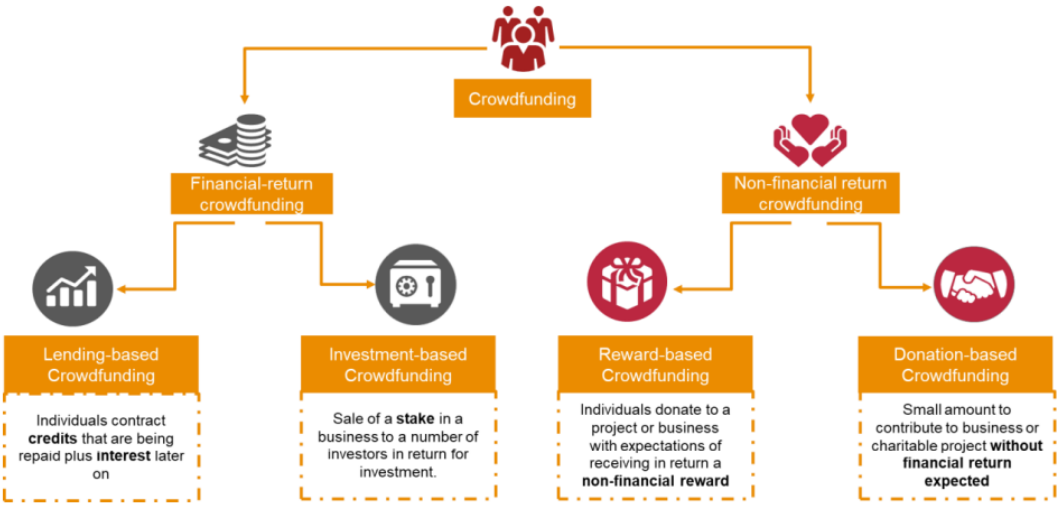
Crowdfunding, as defined by Mollick (2014, p. 2), "refers to the efforts by entrepreneurial individuals and groups - cultural, social, and for-profit - to fund their ventures by drawing on relatively small contributions from a relatively large number of individuals using the internet, without standard financial intermediaries". Similarly, Belleflamme et al. (2015) describe crowdfunding as an open call for funding that typically occurs online through crowdfunding platforms, where contributors may receive different types of returns. The emphasis on "open call" highlights a significant difference from traditional funding. Instead of negotiating with a limited number of decision-makers, creators turn to the general public. Crowdfunding is a subtype of crowdsourcing that refers to the act of outsourcing specific tasks to the crowd through an open call online (Kleemann et al., 2008). According to a third definition, it is a form of online financing, mainly for innovative companies, projects, and individuals, which can take many different forms and serve multiple different purposes, but the common denominator is the financing process that takes place on crowdfunding platforms, which brings together supporters and recipients of support, i.e., backers and creators (Wenzlaff & Gumpelmaier, 2018). These definitions have become fundamental in academic literature, as they highlight the key characteristics of crowdfunding while having multiple elements in common. Most importantly, crowdfunding relies on the crowd, a large number of geographically dispersed backers who typically make relatively small financial commitments. Moreover, it is predominantly internet-based and online platforms like Kickstarter or Indiegogo act as intermediaries between both sides of the market. Using these platforms helps to remove the reliance on traditional financial institutions, and makes it more inclusive and open to anyone. In addition to raising capital, it offers creators numerous additional benefits such as validating market demand, building an initial community of supporters, improving brand visibility and feedback, and enabling open innovation processes by allowing creators to interact and engage with a diverse and knowledgeable crowd that provides ideas, feedback, and early user insights throughout the development process (Chu et al., 2019; Stanko & Henard, 2017).

2.1.1. Crowdfunding models

Crowdfunding models can be broadly divided into two main types, namely "financial-return crowdfunding" and "non-financial return crowdfunding" (European Commission, 2021). Within this framework four main types of crowdfunding can be distinguished: lending-based, investment-based,

reward-based, and donation-based crowdfunding. Financial-return crowdfunding consists of lending and investment-based models. In these models, investors expect a financial return, usually in the form of interest or equity (Belleflamme et al., 2015). However, reward- and donation-based crowdfunding are regarded as non-financial return models. These types of crowdfunding are characterized by non-monetary rewards or purely altruistic motives (Gerber & Hui, 2013), as shown in Figure 1.

Figure 1. The different types of crowdfunding
Source: European Commission, 2021



2.1.2. Crowdfunding Participants

Crowdfunding involves several actors who are interacting in a digital platform ecosystem (Hein et al., 2020). The key players are the platform, the project creators and the backers. Each one plays a unique role in shaping the dynamics and outcomes of campaigns. Project creators, among other things, initiate their campaigns with showcasing their ideas, defining the funding goal, campaign duration and pledge levels. They are responsible for creating the content, including textual descriptions, images and videos that are crucial communication tools for attracting potential supporters. Besides starting the campaign, creators are actively managing the communication by giving updates and answering comments that can strongly influence the perceived credibility and reliability. Backers are individuals who financially support campaigns. In reward-based crowdfunding, they usually receive non-monetary rewards in return for their financial contributions. These non-monetary rewards can be early access to products or exclusive products. Their motivations usually can be explained by a combination of intrinsic and extrinsic drivers, such as an interest towards the product, a desire to support innovation, community belonging or altruistic intentions

(Gerber & Hui, 2013). Backers provide financial support but also social proof and momentum effects (Colombo et al., 2014; Kuppuswamy & Bayus, 2018). Crowdfunding platforms act as intermediaries in a two-sided market that connects creators and backers and reduces transaction costs (Rochet & Tirole, 2003). They provide the technical infrastructure with campaign hosting, payment processing, and tools to boost visibility. They also establish rules and guidelines, such as funding models, fee structures and content guidelines. Platforms can influence the visibility of projects and thereby, the success of campaigns with algorithmic rankings, category structures or featured project sections, such as "Projects We Love" on Kickstarter (Mollick, 2014). The interactions between these three players create a digital marketplace characterized by information asymmetries, trust-building mechanisms and social influence dynamics (Agrawal et al., 2014; Colombo et al., 2014; Vismara, 2018). By understanding these roles of participants, the determinants of success of crowdfunding can be analyzed.

2.1.3. Campaign Lifecycle

Crowdfunding campaigns have a well-defined life cycle with a number of phases, from pre-campaign to post-campaign fulfilment. Reward-based crowdfunding projects generally have three phases: pre-launch, active funding period and post-campaign phase (European Commission, 2021), although the exact process may differ between platforms. In the pre-launch phase the creators prepare the campaign. The success or failure of a project can be influenced by decisions made by creators at this stage, such as setting of the funding goal and the duration, selecting reward options, etc. The active funding period begins right after the campaign has been posted and launched on the crowdfunding platform. Backers are able to pledge financial support within a limited time window during this phase. Research has shown that the dynamics of funding often follow a U-shape pattern (Kuppuswamy & Bayus, 2018). At the beginning, usually there is a rapid and significant inflow of capital that is likely the effect of initial excitement and novelty as well as the support of the creator's network. This is often followed by a slowdown around the middle of the campaign, then, as the deadline is approaching, there is often a last-minute rush that is driven by a sense of urgency, social proof effects and the fear of missing out. The post-campaign phase begins after the financing period ends. If the campaign is successful, the creators are responsible for providing the promised rewards. This phase may affect the campaign's long-term reputation and trust, but it does not directly affect the result of the campaign.

2.1.4. Risks in Crowdfunding

Crowdfunding offers unique benefits to campaign creators and backers, but it also involves significant risks for all participants. One of the most common problems with crowdfunding projects is project failure, when campaigns fail to reach their funding goals or cannot fulfil their promises. Statistics show that 58.02% of Kickstarter campaigns fail as of January 2025, meaning that most projects do not raise sufficient funds (Statista, 2025a). Even when campaigns reach their funding goals, delays in production or delivery are common, particularly in reward-based crowdfunding, where products have not yet been developed and information is limited (Belleflamme et al., 2014; Mollick, 2014; Kim et al., 2017; Shneor et al., 2021). Moreover, operational risks such as production, supply chain and logistics may affect the quality and timing of deliveries, and thus may reduce backer satisfaction (Belleflamme et al., 2014). Such problems are particularly prevalent with tangible goods. Unexpected delays or quality problems may lead to a significant loss of trust and satisfaction of the backers (Kim et al., 2017). Research indicates that approximately 75% of successfully funded Kickstarter projects face significant delivery delays and 9% of them fail to deliver the promised rewards (Mollick, 2014; 2015). Further research on product development campaigns found that only 32% were on time, while 43% were more than a year late (Jensen & Özkil, 2018). Crowdfunding suffers from a large information asymmetry as the creators of the campaigns know considerably more about the feasibility and details of the projects than the backers who provide the funding (Belleflamme et al., 2014; Agrawal et al., 2014; Wessel, 2016; Wessel et al., 2021). This imbalance contributes to uncertainty and raises the possibility of adverse selection (Akerlof, 1970). This issue can be intensified through herding behaviour, where initial backers' contributions are interpreted as signs of credibility (Zhang & Liu, 2012). Rational herding can enhance the momentum of funding (Vismara, 2018), but it can also generate information cascades that can lead to sub-optimal funding decisions in case of inaccurate or misleading information (Agrawal et al., 2014; Colombo et al., 2014). To help reduce these risks, creators should use clear and visible signs, observable cues, in accordance with signalling theory to demonstrate their project is of high quality and trustworthy (Spence, 1973). Another problem in crowdfunding is the "Meager Funding-Target" in which creators set small funding targets to increase the probability of meeting the "all-or-nothing" requirement and achieving campaign success (Wessel et al., 2021). Although this approach might improve the probability of reaching the funding goal, it often causes a lack of resources that could eventually decrease backer satisfaction (Wessel et al., 2021). Campaign organizers also risk reputational damage and legal compliance. Failure of a project or delay in delivery of rewards can damage the reputation of the creator and reduce

the creator's ability to attract future support (Mollick, 2014). Moreover, all platforms have their own regulatory requirements, which may result in legal issues for creators and supporters in terms of intellectual property, taxes or consumer rights (Dushnitsky et al., 2016). All these risks illustrate the complexity of crowdfunding and show that many factors can influence the outcome, which makes it difficult to predict success and underscores the need for careful evaluation.

2.1.5. Success Factors and the Definition of Success in Crowdfunding

Success in crowdfunding is not a uniform concept, it is closely tied to the underlying financing model. Crowdfunding campaigns usually adopt either a Keep-It-All (KIA) or an All-Or-Nothing (AON) model. Cumming et al. (2015) explained that in a KIA model, the campaign creator can keep the entire raised amount, even if the funding goal hasn't been achieved, whereas in the AON model, the creator only receives the collected funds if the campaign reaches or exceeds the predetermined funding threshold. The AON model thus serves as a provision point mechanism, which mitigates several risks inherent in crowdfunding environments (Agrawal et al., 2014). Specifically, it solves the free-rider problem, serves as a credible signal of project quality and reduces information asymmetry and, therefore, reduces perceived risks for backers and increases trust in the campaign. Overall, AON campaigns tend to be more successful than KIA campaigns, as these mechanisms improve coordination among backers, encourage early funding momentum, and increase the likelihood that campaigns will reach their funding goals (Mollick, 2014; Cumming et al., 2015). Kickstarter campaigns follow the AON model, thereby, in this study, success is defined as achieving or exceeding the funding goal during the specified duration of the campaign.

It is important to understand the factors leading to success in crowdfunding, which have been a central element of research for many years. Building trust is known as a key objective in online marketplaces, which supports the foundations of e-commerce, digital marketing and development of online communities (Shneor et al., 2021). This is particularly relevant for reward-based crowdfunding campaigns where ambiguity is higher and information is less available. Therefore, the success of projects largely depends on the design and management of campaigns by the creators and how they can effectively signal credibility, reduce uncertainty, and build trust. There are many studies on what factors lead to success and what makes a campaign successful. For instance, project descriptions are very important because they provide the potential backers with the key information they need. Research shows that having a clear, well-written description and good quality visuals can help a campaign succeed (Mollick, 2014; Koch & Siering, 2015; Abuamer et al., 2025). Moreover, Shneor

et al. (2021) found that studies have identified a positive correlation between success and sophisticated language use. Song et al. (2019) for example, found that in the games category, the inclusion of certain words in the title, such as "visual" and "novel", increases their likelihood of success. Additionally, it has been also found that the number of reward options has a positive impact on success (Mollick, 2014; Shneor et al., 2021; Lin & Boh, 2021; Li et al., 2022). On the other hand, too many reward options can have the opposite effect, and it can actually decrease backer participation, creating a "Goldilocks-type" effect (Elitzur et al., 2024). It has been also proven that campaigns with lower funding goals are more likely to succeed (Mollick, 2014; Baber, 2020; Lin & Boh, 2021; Shneor et al., 2021; Elitzur et al., 2024), because higher goals are often associated with higher perceived risk and complexity (Koch & Siering, 2015). Furthermore, it is a misconception that projects with longer durations can win over more backers. Song et al. (2019) revealed that the longer the duration of the projects is, it is more likely to fail. Shorter durations, specifically between 10 to 15 days, are proven to be most effective for maximizing success probability (Elitzur et al., 2024). Yancey Strickler, the co-founder of Kickstarter, also confirmed that a longer campaign period does not necessarily have a positive effect on its success, as it makes it easier for supporters to procrastinate, who may even forget to return to the campaign (Song et al., 2019). Aligning a campaign with popular project categories or launching it during high-activity periods is also identified as an external factor for success (Abuamer et al., 2025). Baber (2020) also concluded that worldwide shipping of rewards positively influences success and creators who have a well-proven track record and good reputation have higher chances for a successful campaign. Beyond static campaign characteristics, many dynamic factors positively affect campaign outcomes during the funding period. The early support of backers is significantly important, because as funding accumulates, it can trigger herding behaviour and encourage more people to invest (Agrawal et al., 2014). According to Baddeley (2010, p. 282), "Herding can be defined as the phenomenon of individuals deciding to follow others and imitating group behaviours rather than deciding independently and atomistically on the basis of their own, private information." This phenomenon is a response to uncertainty and the inherent limitations of human knowledge (Keynes, 1930). This "rational herding" occurs as potential investors use the funding decision of others as a credible signal of project quality, leading to an increased willingness to invest as the project approaches its goal (Agrawal et al., 2014; Alegre & Moleskis, 2016). However, as noted by Belleflamme et al. (2015), backers are usually more satisfied with the project they support if there has been no irrational herd mentality. As herding can lead to increased funds, it can also induce information cascades which trigger a path-dependent process,

where the ultimate success of a campaign depends heavily on the quality of screening performed by early funders (Agrawal et al., 2014). Accumulated funding can become an unreliable indicator of project quality and potentially lead to suboptimal outcomes and lower backer satisfaction if these initial signals are weak or biased. Furthermore, giving regular updates signals that the creator is committed to the project. Signs of limited availability, including early bird rewards or exclusive products, also increase funding by creating a sense of exclusivity and "pursuit of pleasure" among backers (Thürridl & Kamleitner, 2016; Kunz et al., 2017).

2.2. Classical Machine Learning Models

Classical machine learning models continue to be widely used in predictive analytics due to their interpretability, computational efficiency, and strong theoretical background (James et al., 2021). These models are generally more effective on structured tabular data than more complex deep learning methods. Since crowdfunding datasets typically consist of structured campaign characteristics (e.g., funding goal, duration, category, textual features), classical methods provide a suitable and robust baseline for predictive modeling. In this study, several supervised classification models are considered, where the primary objective is to predict campaign success as a binary outcome variable.

2.2.1. Logistic Regression

Logistic regression (Cox, 1958) is a supervised machine learning algorithm that is often used for binary classification tasks (Hosmer & Lemeshow, 1989). It estimates the probability that a given instance belongs to a particular class, for example, the probability that a particular Kickstarter campaign will be successful or not (Géron, 2022). Similarly to Linear Regression, first it computes a weighted linear combination of the input features and a bias term. However, Logistic Regression does not provide a continuous output but uses the logistic (sigmoid) function (Schumacher et al., 1996) to map the results to a probability between 0 and 1. In a formal way, we can model the probability of success using the logistic (sigmoid) function:

$$\mathrm{P}(\mathrm{y}=1 \mid \mathrm{X})=\sigma(\beta_0+\beta_1 X_1+\cdots+\beta_\mathrm{n} X_n)=\frac{1}{1+e^{-(\beta_0+\sum \beta_i X_i)}}$$

In this, β_0 is the intercept, and $\beta_{1,\ldots,}\beta_\mathrm{n}$ are the coefficients of the predictor variables. This transformation guarantees that the predicted probabilities are in the interval of $[0,1]$.

Logistic Regression is highly interpretable because the estimated probabilities can be transformed into probability ratios (Schumacher et al., 1996). Probability ratios show how the odds of an outcome change if we increase a specific predictor variable by one unit, holding all other variables constant (Hosmer & Lemeshow, 1989). Logistic Regression is thus especially useful for research purposes where knowing the direction and magnitude of effects is also very important (James et al., 2021; Hosmer & Lemeshow, 1989). However, the model assumes linearity between independent variables and the log-odds of the dependent variable. It is a robust baseline that performs surprisingly well when relationships are approximately linear (James et al., 2021).

2.2.2. Support Vector Machine (SVM)

Support Vector Machine is another fundamental machine learning algorithm that is often used for both regression and classification tasks. Vapnik and Chervonenkis (1963) laid the foundations of this model in statistical learning theory and Boser et al. (1992) and Cortes & Vapnik (1995) further developed it. It tries to maximize the distance between different classes by finding an optimal decision boundary, which is called hyperplane (Vapnik & Chervonenkis, 1963; Boser et al., 1992). The hyperplane in a *d*-dimensional space is described by the following linear function:

$$f(x) = w^T x + b$$

where w is the weight vector and b is the bias term. This margin is defined as the distance between the hyperplane and the nearest data points from each class. They are called support vectors, which are to be maximized. In practice, specifically by minimizing $\frac{1}{2} \| w \|^2$ the model aims to improve generalization and reduce classification errors in unseen data.

If the data is not linearly separable, the kernel trick is used to map the original data into a higher dimensional space where it eventually becomes linearly separable (Aizerman et al., 1964; Boser et al., 1992). Common kernels are linear kernel, polynomial kernel, and radial basis function (RBF) kernel. These kernel functions enable SVMs to implicitly learn complex, nonlinear dependencies without explicit calculation of high-dimensional transformations. SVM is very well suited to high dimensional data, where data is characterized by a large number of features and relatively few samples (Cortes & Vapnik 1995). However, the algorithm can be computationally expensive for very large datasets and requires careful hyperparameter tuning, such as the regularization parameter (C) and kernel-specific parameters (James et al., 2021). In binary classification tasks, such as crowdfunding success prediction, SVM assigns class labels based on the position of the observations relative to the learned hyperplane, typically using the sign of the decision function: $y = \text{sgn}(w^T x + b)$.

2.2.3. Decision Tree

Decision Tree is a supervised machine learning algorithm that is used for classification and regression problems. It is based on a hierarchical decision making process that recursively partitions a dataset into smaller subsets based on feature values (Breiman et al., 1984). The algorithm was developed by Breiman et al. (1984) and popularized in the framework of CART (Classification and Regression Trees). The decision tree model constructs a tree of internal decision nodes, branches and final leaf nodes (Breiman et al., 1984). The algorithm chooses the feature and threshold value at each internal node that most effectively separates the classes into homogeneous subsets. The quality of the partition is often evaluated in classification tasks using purity measures such as the Gini index or entropy (Breiman et al., 1984; James et al., 2021). The tree-building process is recursive and continues until a stopping criterion is met. The stopping criteria can be maximum tree depth, minimum number of samples per leaf or minimum impurity improvement (Breiman et al., 1984). The final prediction is obtained by following the decision path from the root to the leaf node.

Decision trees have many advantages. The model is easy to interpret because the decision rules are visualized and easy to understand (James et al., 2021; Géron, 2022). It can capture nonlinear relationships between variables and does not require feature scaling and/or normalization (James et al., 2021). However, the individual decision trees tend to overfit especially when the tree grows deep and becomes too complex (James et al., 2021; Géron, 2022). Although this model is relatively simple, it is considered to be the backbone of more advanced ensemble methods like Random Forest and boosting algorithms.

2.2.4. Random Forest

Random Forest is a decision-tree based ensemble learning method that is also widely used for classification and regression tasks (Breiman, 2001). It is a bagging (Bootstrap Aggregating) method, that was originally proposed by Breiman (1996), which aims to improve predictive performance while reducing overfitting compared to a single decision tree. The model builds a large number of trees during training, and all trees are trained on a bootstrap sample of the original dataset, which means that the observations are sampled randomly with replacement (Breiman, 2001). Additionally, only a random subset of features is considered for each split of the tree. This additional randomness removes the correlation between trees and makes the model more robust (Breiman, 2001). In this case, all trees predict the outcome, and the final decision is defined by the majority vote of all trees for classification

tasks (Breiman, 2001). The predicted probability of success is equal to the proportion of trees that rate the campaign as successful.

Random Forest has many advantages. First of all, it is able to capture complex, nonlinear relationships between variables without requiring them to be specified in advance (Hastie et al., 2009). It is relatively robust to outliers and multicollinearity, and effectively handles high-dimensional datasets, which is particularly useful when one-hot encoding or textual feature extraction is used, as in this study (Biau & Scornet, 2016).

2.2.5. Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting is a state-of-the-art ensemble learning algorithm that is based on gradient boosting decision trees. The method was introduced by Chen and Guestrin (2016) and it has become one of the most utilized machine learning algorithms for structured data prediction. Boosting algorithms build models sequentially, unlike bagging approaches (e.g. Random Forest) (Schapire, 1990; Freund & Schapire, 1997; Hastie et al., 2009). In boosting, each new tree is trained to correct the errors of the previous ensemble tree. This implies that the following trees focus on the observations that were misclassified before and continuously improve the accuracy of the model (Hastie et al., 2009).

XGBoost has adopted several optimization methods to improve the traditional gradient boosting. The model minimizes a regularized objective function that is a combination of a differentiable loss function and a complexity penalty term (Chen & Guestrin, 2016). This regularization term is used to control the model complexity and reduce the risk of overfitting. XGBoost also uses the second-order Taylor expansion of the loss function that enables more precise optimization during the tree construction process (Chen & Guestrin, 2016). XGBoost is also highly scalable and suitable for large datasets, because it effectively handles missing values, uses parallelized tree learning and has optimized memory usage. In binary classification problems, such as the prediction of crowdfunding success, XGBoost produces probability estimates by combining the predictions of the boosted tree ensemble. The final prediction is usually obtained by applying a classification threshold to the estimated probability. It has many advantages as it is able to model complex, nonlinear relationships and high-order feature interactions. In general, the algorithm makes strong predictions and has built-in regularization mechanisms, which improve the generalization ability. However, XGBoost requires careful hyperparameter tuning, including parameters such as learning rate, maximum tree depth, and number of boosting rounds (Chen & Guestrin, 2016; James et al., 2021).

2.3. Deep Learning Models

2.3.1. Neural Networks

Artificial neural networks (ANNs) are computational systems inspired by the human brain (McCulloch & Pitts, 1943). They process complex data patterns through multiple layers of interconnected nodes, known as neurons, that enable the network to automatically learn hierarchical features (Rumelhart et al., 1986). Each neuron takes several inputs, computes a weighted sum, passes it through a nonlinear activation function, and then forwards the output to the following layer (Goodfellow et al., 2016). In a neural network there are normally three different types of layers. The input layer takes in the raw features from the dataset, where each node corresponds to a single input variable (Bishop, 2006). Then the hidden layers do intermediate computations and transform the input by applying learned weights and biases and finally apply a nonlinear activation function (Goodfellow et al., 2016). Deep neural networks have multiple hidden layers to extract more abstract and complex representations from the data (LeCun et al., 2015). The output layer is responsible for generating the final prediction, and its structure depends on the nature of the problem. For instance, a single neuron with sigmoid activation is commonly used for binary classification, whereas multiple neurons with Softmax activation are employed for multiclass classification tasks (Goodfellow et al., 2016). The operation performed by a neuron can be mathematically expressed as follows:

$$y = f\left(\sum_{i=1}^{n} w_i x_i + b\right)$$

Where x_i are input features, w_i are weights, b is the bias term, and $f(\cdot)$ is a nonlinear activation function. Some common activation functions include ReLU, Sigmoid and Tanh (see table 1). Swish is a smooth non-monotonic function defined as $f(x) = x \cdot \sigma(x)$ which can outperform ReLU in deep learning. Unlike ReLU, which cuts off all negative inputs, Swish allows small negative values to pass that improves the gradient flow and hence optimization in deeper networks (Ramachandran et al., 2017). Neural networks can learn more complex relationships than just linear boundaries through the introduction of nonlinearity (Goodfellow et al., 2016).

Table 1. Popular nonlinear activation functions

Function	Formula	Range	Key Characteristic
Sigmoid	$\sigma(z) = \frac{1}{1+e^{-z}}$	(0,1)	Maps output to probabilities, smooth gradient, output layer for binary classification
Tanh	$\text{Tanh}(z) = \frac{e^{z}-e^{-z}}{e^{z}+e^{-z}}$	(-1,1)	Zero-centered, stronger gradients than sigmoid, rarely used in hidden layers
ReLU	$f(z) = \max(0, z)$	(0,∞)	Computationally efficient, solves vanishing gradient, standard choice for hidden layers
Leaky ReLU	$f(z) = \max(\alpha z, z)$	(-∞,∞)	Prevents "dying neurons" by allowing small negative values
Swish	$f(z) = z \cdot \sigma(z)$	≈ -0.122, ∞)	Smooth, non-monotonic, improves gradient flow
Softmax	$\frac{e^{z_i}}{\sum e^{z_j}}$	(0,1)	Probabilities sum to 1 across all classes, output layer for multi-class tasks

Neural networks are trained by adjusting their weights to minimize a loss function $L(y, \hat{y})$ that measures the difference between the networks predictions $\hat{y}$ and the actual target values y (Bishop, 2006). The learning process generally involves backpropagation and gradient descent. During the training, the network calculates the gradient of the loss function with respect to each weight and updates the weights iteratively in order to find the global minimum of the loss function (Rumelhart et al., 1986):

$$w_i \leftarrow w_i - \eta \frac{\partial L}{\partial w_i}$$

where η is the learning rate. The network identifies patterns in the training data through this iterative process and continuously updates its parameters to improve its performance (Goodfellow et al., 2016). Modern ANNs often employ additional strategies to improve their ability to learn and generalize. Regularization methods such as dropout functions by randomly disabling certain neurons during training, which helps to reduce overfitting (Srivastava et al., 2014). Batch normalization, meanwhile, normalizes the input to each layer, thereby facilitating faster convergence (Ioffe & Szegedy, 2015). Advanced optimization approaches like Adam and RMSprop dynamically adapt the learning rate throughout training, enhancing both stability and efficiency (Kingma & Ba, 2014; Tieleman & Hinton, 2012). These methods are often combined with scheduling the learning rate (e.g. ReduceLROnPlateau) in order to further fine-tune the optimization process by reducing the step size when the loss starts to stagnate (Bengio, 2012). Furthermore, early stopping can be used to prevent overfitting by stopping training when performance on validation data no longer improves (Prechelt, 1998).

Depending on the type of input data neural networks can be adapted to different specialized architectures. For instance, convolutional neural networks (CNNs) are great for processing visual and spatial data (LeCun et al., 1998), while recurrent neural networks (RNNs) and transformer models are better for handling sequential or temporal data, like text, time series, or progress updates in crowdfunding campaigns (Hochreiter & Schmidhuber, 1997; Vaswani et al., 2017). Neural networks are very useful for predictive analytics tasks like predicting the outcome of crowdfunding campaigns, especially when the datasets are high-dimensional or have complex relationships between variables (Bengio et al., 2013). Although these models generally require increased computational resources and careful tuning of hyperparameters compared to more conventional methods (James et al., 2021), they are able to detect nonlinear patterns and complex interactions. This makes them very efficient in extracting valuable insights from large and structured data. Neural networks are also good at processing multimodal information, i.e. combining different data sources such as structured variables, textual descriptions or images (Ngiam et al., 2011). These models can be flexibly designed by having distinct representations for each input type through separate dedicated input branches, which can then be combined in a unified architecture. Each modality is generally mapped to a latent feature vector using modality-specific layers and these representations are fused by fusion techniques such as concatenation or additional fully connected layers (Baltrusaitis et al., 2018). This allows the network to learn modality-specific features and the interaction between different types of data. The choice of fusion strategy in neural network design is determined by the model architecture. In early fusion, the raw features of the different data types are fused prior to processing, allowing the network to learn from combined input sources (Snoek et al., 2005). On the other hand, intermediate fusion fuses representations learned in hidden layers allowing more complex interactions between modalities (Baltrusaitis et al., 2018). Late fusion combines the results of separate models, each trained on a different type of data, and aggregates their outputs to make a final prediction (Snoek et al., 2005).

2.3.2. Large Language Models (LLMs)

Large language models (LLMs) are deep learning models that are developed to understand, interpret, and generate human language (Zhao et al., 2023). These models are usually based on transformer architectures (Vaswani et al., 2017), which use self-attention mechanisms to explore the relationships between words or tokens in a sequence. This allows them to learn both short- and long-range dependencies (Ferraris et al., 2025). Large language models are trained on large corpora of text and are able to learn nuanced and contextual representations of language. They are typically trained with unsupervised tasks such as next-token prediction or masked language modeling. Pre-trained LLMs

can be also adapted later to specific tasks by fine-tuning, where their parameters are updated using task-specific labeled data (Howard & Ruder, 2018; Devlin et al., 2018).

BERT (Bidirectional Encoder Representations from Transformers) is a major large language model that generates deep, bidirectional text representations (Devlin et al., 2018). In contrast to conventional language models that process the text in a single direction, BERT reads the context both before and after the word at the same time, thereby understanding the subtle semantic and syntactic relationships in the text (Devlin et al., 2018). Pre-trained BERT models can be fine-tuned for various natural language processing tasks, such as text classification, named entity recognition or semantic similarity detection, by adding lightweight task-specific layers on top of the frozen or partially trainable encoder (Sun et al., 2020). BERT embeddings can also be extracted and used as dense feature vectors in machine learning applications (Peters et al., 2018).

Large language models are not only just used for traditional natural language processing tasks, but they are also increasingly being utilized for structured classification and prediction problems (Wu et al., 2024). They can learn complex relationships between text and metadata that traditional ML models might not be able to capture. LLMs are also capable of providing comprehensible explanations for their predictions thereby increasing interpretability and transparency over typical "black-box" models (Zhao et al., 2023). Prompt-based techniques such as zero-shot and few-shot learning can also be used for guiding the model by natural language instructions without any additional task-specific training (Brown et al., 2020; Liu et al., 2021). This is especially useful when task-specific datasets are not available or when it is not feasible to fully finetune a model. LLMs can make predictions and provide explanations for new, previously unseen examples by leveraging the knowledge gained from the large-scale pre-training (Zhao et al., 2023). They also perform well on multimodal tasks when structured or visual data are added (OpenAI et al., 2024). While most of the current applications target text, the transformer architecture is able to process different types of inputs, which leads to richer and more detailed representations (Vaswani et al., 2017). LLMs are therefore, very useful for research where both accurate predictions and clear, interpretable results are needed. This enables language understanding to be integrated with predictive modeling and explainable AI.

3. Literature Review

3.1. Success Prediction in Crowdfunding

Identifying and predicting the success of crowdfunding campaigns have attracted significant scientific interest in the last decades. Early studies mainly focused on identifying features that most dominantly drive success (Baber, 2020; Song et al., 2019; Mollick, 2014), whereas more recent works incorporate machine learning and deep learning methods to predict campaign outcome, especially on platforms like Kickstarter. The problem is usually framed as a binary classification task, where campaigns are qualified as successful or failed whether they achieve their funding goal or not.

3.1.1. Traditional Machine Learning Approaches

Most of the existing studies heavily rely on traditional machine learning methods with structured metadata. They usually consider features such as funding goal, campaign duration, category, and creator-related variables and formulate the problem as a binary classification task. These studies usually incorporate post-launch dynamics, such as early funding progress, number of backers over time and social engagement signals that are inherently excellent predictors of campaign outcomes (Zhang et al., 2017; Wang et al., 2020; BaniMustafa et al., 2022; Doula & Chowdhury, 2025). Elitzur et al. (2024), proved that traditional machine learning models are better suited to predict success than traditional regression because they also take complex, nonlinear relationships into account. Several models were implemented and tested in multiple studies, including Support Vector Machines (SVM), Decision Trees, Logistic Regression, K-Nearest Neighbours (KNN) (Greenberg et al., 2013; Yu et al., 2018; Wang et al., 2020; BaniMustafa et al., 2022; Pourroostaei Ardakani et al., 2025).

3.1.2. Deep Learning for Textual and Visual Features

Later studies tried to overcome the limitations of previous work and introduced deep learning methods that are capable of processing unstructured data, especially text and images. According to Tang et al. (2023), in addition to metadata, project descriptions are also a key feature in predicting success because they are well reflecting the perceived quality of the campaign. Therefore, the linguistic features of the campaigns have been studied by many researchers. Wang et al. (2022) analyzed more than 267,000 Indiegogo projects to determine the impact of the readability and clarity of campaign texts on their success. They found that linguistic understandability increases the probability of success and also concluded that lexical indicators such as the number of difficult words are more important predictors than grammatical structure.

Early studies primarily used simple text representation methods such as Bag of Words (BoW) and TF-IDF, which measure how frequently words appear in a text. Other studies applied basic embedding methods such as Word2Vec and GloVe, which generate word representations based on semantic relationships and similarities between words (Mitra & Gilbert, 2014; Kaminski & Hopp, 2020). These techniques are efficient to identify fundamental lexical and semantic trends, but they are not suitable to extract richer contextual relationships and the subtle meanings that words acquire in different contexts. Therefore, to overcome the limitations of these simpler approaches, more recent research has started to adopt more sophisticated embedding models like BERT and FastText (Cheng et al., 2019). They offer a deeper understanding of context because they can capture deep contextual semantics and bidirectional relationships in project descriptions (Gunduz, 2024; Gunduz et al., 2026).

Pictures and videos are also playing pivotal roles in the success of campaigns (Cheng et al., 2019), and many studies started to implement more advanced visual analysis techniques instead of simple image counting to account for this. Blanchard et al. (2023) analyzed the aesthetic features, such as colours and composition, as well as the characteristics of the scene, like the number of identifiable faces. Similarly, Zhang et al. (2021) investigated how cover photo quality and facial attributes relate to project success. Hou et al. (2019) applied deep learning to extract feelings such as sadness and satisfaction that significantly influence backer engagement.

3.1.3. Multimodal Deep Learning Models

As crowdfunding campaigns usually convey values through multiple channels, a few studies recently have even deployed a multimodal approach that integrates metadata, textual, and visual features in one model. For example, Cheng et al. (2019) introduced a multimodal deep learning framework that integrates all three modalities by concatenating it into a fully connected network to predict the success of Kickstarter campaigns on a dataset of 34,112 projects. By combining representations learned from text and images, their multimodal deep learning model significantly outperforms single modal baselines, achieving an F1 score of 0.7534 and AUC 0.8326. This research provides early evidence that the success of crowdfunding is inherently multimodal and that joint representation learning is favourable.

Kaminski and Hopp (2020) further advance multimodal analysis by analyzing textual, speech and video signals in crowdfunding campaigns. They converted text into GloVe representations and used the ResNet architecture to extract features from video data and finally fed them into six linear and nonlinear classifiers, including a Multi-Layer Perceptron. They found that a simple Logistic

Regression (F1=0.71) can outperform more complex neural networks or XGBoost when predicting success from text and that each modality contributes to predictive performance independently. Text is especially important, but the fusion of all modalities does not lead to significantly better results. It is worth noting that no modality is dominating which highlights the need for integrated multimodal models.

Tang et al. (2023) also model specifically the cross-modal interactions between textual descriptions and project videos. They propose a Deep Cross-Attention Network (DCAN) that uses stacked cross-attention blocks that allow the features to guide the selection of the representation and aggregation of the other. In contrast to previous multimodal approaches that usually rely on the concatenation of features, their model also learns finer relationships between modalities, despite the lack of temporal alignment or frame-level annotation. The results show that DCAN consistently outperforms strong unimodal baselines and earlier multimodal models with an F1 score of 0.7312 and at the same time improves interpretability. Since attention mechanism assigns weights to specific keywords and keyframes, it allows the model to visually highlight influential features in prediction.

Al-Qershi et al. (2022) examined the visual and speech features of video with text. They found that ensemble models can achieve an accuracy of 82%. The study also showed that campaigns are more likely to succeed when they are more persuasive and for example include people in videos, use a positive tone, and share personal experiences. Bao et al. (2025) introduced a theory-driven multimodal framework that combines textual, image, and video data and uses systematic functional theory to guide feature creation. Instead of relying solely on data-driven representations, their approach captures three key communication dimensions: ideational, interpersonal, and textual. Thus it connects multimodal signals to the underlying communication processes. Their results show that multimodal models significantly outperform unimodal approaches and that combining theoretically informed features further improves predictive performance. This study not only emphasizes the role of multimodal integration but also proves that using theory-driven concepts can boost model performance and interpretability.

Cai et al. (2024) further improved the multimodal prediction by proposing a Multimodal Dynamic Graph Convolutional Network (MDGCN). This model considers campaigns as nodes in a dynamic graph, leverages textual, visual and metadata attributes, and accounts for all temporal dependencies, including early funding dynamics and social interactions. It learns the modality-specific patterns and structural relationships by applying graph convolutions on the temporal aspect and outperforms

traditional baselines and static multimodal models. It highlights the essential role of cross-modal interactions and temporal dynamics in predicting success.

Abuamer et al. (2025) conducted a comprehensive analysis of deep learning architectures on a Kickstarter dataset with 482,695 projects. Their results showed that neural architectures, especially Bidirectional LSTMs (BiLSTM) are consistently better than traditional machine learning methods because they capture the complete contextual patterns in both forward and backward directions. However, the authors pointed out that deep learning models are more accurate (93% in their study) but also increasingly complex and less interpretable, especially in multimodal settings. To overcome this "black box" effect, some subsequent studies have combined SHAP (Shapley Additive exPlanations) values and information processing theory (IPT) to provide a more rigorous explanation on how each characteristic affects funding outcomes (Yang et al., 2025; Tang et al., 2026). Similarly, Cavalcanti et al. (2024) highlighted the use of SHAP in recent crowdfunding research, showing how it can identify the most influential features and maintain high predictive performance.

3.1.4. Large Language Models and Emerging Reasoning-Based Approaches

Although LLMs have been widely studied in the field of language comprehension and generation, their application in structured prediction tasks with explanation generation is still underexplored in the field of crowdfunding. Existing research focuses on learned representations and performance metrics and disregards qualitative reasoning completely or treats it as secondary. LLMs provide a unique advantage in this sense as they are able to provide natural language explanations besides predicting success, which can potentially bridge the gap between quantitative forecasting and qualitative reasoning. For example, Angus et al. (2025) developed CrowdInsight, which uses a multimodal neural network with GPT-4o and SHAP values to generate explanations for predictions. Bhaumik et al. (2025) also demonstrated that GPT-4o can pick up subtle linguistic, social, and emotional elements of campaign narratives, such as empathy, urgency, and perceived financial vulnerability. If they are combined with traditional machine learning they can also measure how much they contribute to the outcome. Similarly, Schoenegger et al. (2024) showed that an ensemble of 12 LLMs can evaluate live campaigns, outperforming human crowds, while providing coherent reasoning about which projects demonstrate more compelling multimodal content. These studies illustrate that LLMs are capable of not only predicting campaign success but provide coherent, human-readable reasonings for their decisions.

3.2. Research Gap

Crowdfunding continues to grow as an important financing mechanism for entrepreneurs, creatives, and startups, making it highly relevant to identify and predict campaign success. Although previous studies have made significant progress in this area, a lot of limitations still remain. First of all, much of the literature focuses on either structured metadata or textual content (or the combination of both), and rarely incorporates images that are a crucial part of campaign presentation. As a result, the richness of crowdfunding campaigns is often not fully exploited. This research aims to evaluate whether multimodal learning improves predictions of crowdfunding success by integrating textual, visual, and structured features into a single framework. Moreover, traditional machine learning and deep learning approaches usually focus on accuracy, while interpretability and explainability methods are often quantitative and fragmented across different model types. This limits the usability from a human perspective for understanding and decision-making. The goal of this study is to bridge this gap by incorporating explainability techniques and feature importance methods, as well as utilizing LLMs to predict and provide qualitative reasonings as well. Thus, this enables a better understanding of the most influential features that drive success. Additionally, existing studies also often use post-launch dynamics, which substantially improve predictive performance. However, in real-world settings it is highly desired that project outcome can be predicted before launch, when creators are still able to modify campaign design and strategy. This research aims to develop models that support early-stage decision-making by focusing only on pre-launch information and thereby increasing practical relevance. These considerations have multiple practical implications for several stakeholders. Creators would be able to assess the feasibility of their campaigns before launching and make informed modifications based on the feedback. Platforms, such as Kickstarter, would be able to better identify promising campaigns in early stages and therefore improve their recommendation systems and quality controls. Additionally, understanding the drivers of success helps creators to design more effective campaigns. Therefore, building a model that is capable of predicting crowdfunding success and revealing stronger and weaker features prior to launching would result in a particularly valuable tool for campaign creators. From a data science perspective, the project addresses a broader methodological challenge on how to effectively integrate heterogeneous multimodal data sources in a way that improves predictive performance and also produces interpretable insights. This paper aims to provide a comprehensive assessment of predictive performance, interpretability and quality reasoning based on experimenting with LLM-based approaches, traditional machine learning, and multimodal deep learning methods in multiple input

configurations. This makes this study not only relevant in the field of crowdfunding but also a valuable contribution to applied deep learning research.

4. Methodology

4.1.Data Source and Context: Kickstarter Platform

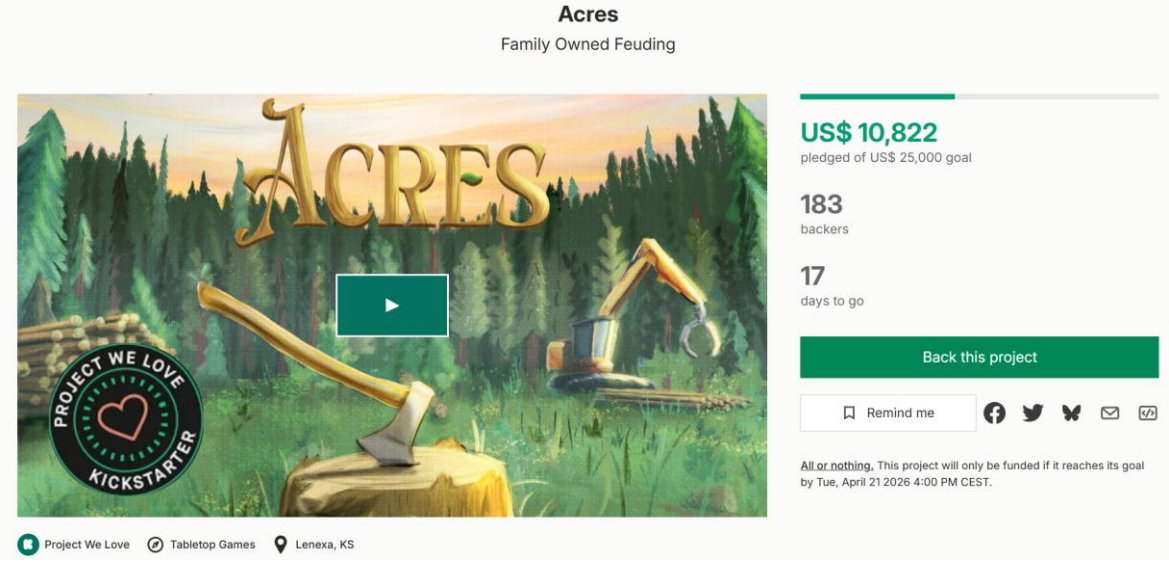
Kickstarter is one of the biggest and best-known reward-based crowdfunding platforms (Song et al., 2019), where creative projects can be backed across a total of 15 categories, from movies to games, music, art, design and technology among others. It launched in 2009, as of August 2025, it has attracted 24.47 million backers since, of which around 8.53 million are returning (Statista, 2025c). It has raised over $8.99 billion in total, hosting over 650,000 campaigns with a success rate of 41.98% (Statista, 2025b). Currently people can launch campaigns from 25 countries, while projects can be backed from anywhere in the world. This research uses Kickstarter campaigns, as it is a widely used platform and has extensive publicly available data.

4.1.1. Structure of a Kickstarter Campaign Page

Kickstarter campaign pages are standardized to be accessible and easy to navigate for backers. While creators have freedom in presenting their projects, the pages include consistent elements that ensure comparability and transparency.

Figure 2. Project header of an example campaign on Kickstarter

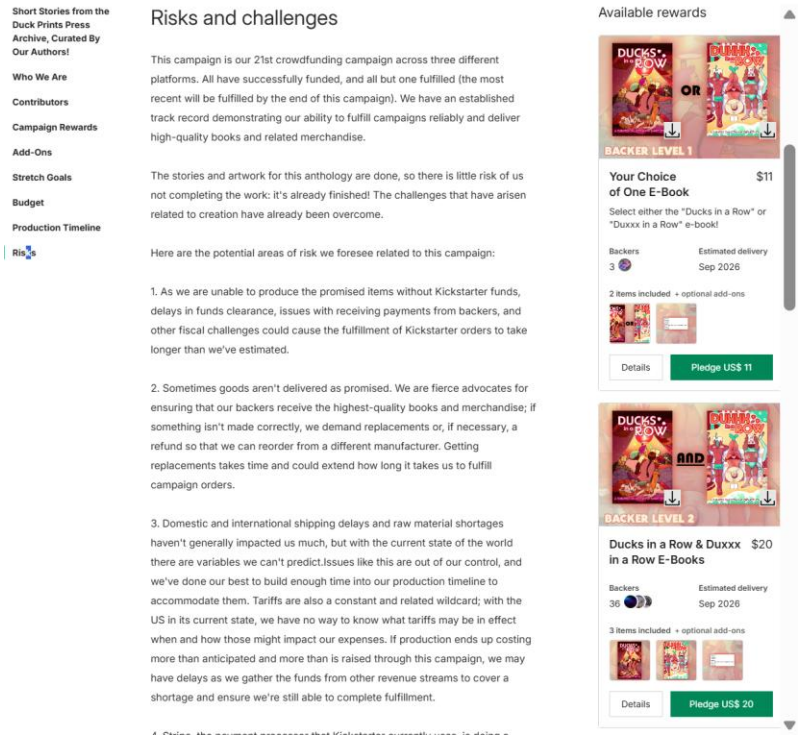
Source: McWhirter, 2026



The project header on the top of the page contains fundamental, high-level information (Figure 2). This usually includes the project title, a short description of the idea called blurb, the project category, location and the name of the creator, as well as a title image or video. Moreover, the header highlights the funding goal, the amount pledged, the number of backers, and if the project is live, it shows the time remaining until the deadline, thus making the progress of the campaign transparent. This creates a sense of urgency and social proof that can influence the behaviour of potential backers (Colombo et al., 2014; Kuppuswamy & Bayus, 2018). The main body of the campaign page contains the detailed description of the project, which is also called story. Here, the creators can thoroughly explain their ideas and highlight the features of their product. Here, they often combine text with visual features and incorporate many images, videos, or any embedded media to capture attention and improve engagement. This section plays a central role in mitigating information asymmetry since it provides the most important information that backers would need to be able to assess the feasibility and value of the campaign. All project pages have a "Risks and Challenges" section at the bottom, where creators can discuss the risks related to manufacturing, shipping, or fulfilment, and explain how they plan to mitigate them (Figure 3). This section is officially present on all campaign pages, however, it is not compulsory to fill it in. As a result, the published content can influence the perceived transparency and reliability, as it can indicate that the creators can realistically address uncertainties.

Figure 3. Risks and rewards sections of an example campaign on Kickstarter

Source: Duck Prints Press, 2026



Another fundamental part of a campaign page is the rewards section that is usually beside the campaign description (Figure 3). It presents the different pledge levels available to backers. Each pledge level has an associated minimum pledge amount and rewards, like early access, physical products, or symbolic rewards. This section connects financial support to tangible or intangible rewards and plays a key role in shaping the willingness of backers to support. The campaign page also includes a small section about the creator, which is usually displayed above the rewards section. This includes a brief introduction and indicators of prior experience, such as the number of created as well as backed projects. Furthermore, Kickstarter campaign pages also include interactive and social features, such as comments, updates and sharing options. These are usually not integrated into the main page but available on subpages. They enable the communication between creators and backers and allow potential backers to follow ongoing interactions and facilitate the spread of campaigns through social networks. Together, these interactive features contribute to community building, further reducing uncertainty and increasing trust in the project.

4.2. Data Collection

The primary dataset used in this analysis was provided by WebRobots.io. It is publicly available and built by a scraper robot that crawls all Kickstarter projects monthly since 2014. 83 individual CSV files were obtained that were combined into a unified dataset. Each file contained several fields with nested JSON structures (e.g., project creator, profile, location) that were flattened, and faulty or malformed JSON entries were deleted. As a result, the initial dataset contained 257873 rows and 110 features. These features included mostly project-level metadata such as title, category, goal amount, pledged amount, currency, etc.

4.3. Dataset Optimization & Data Enrichment by Web Scraping

Initially, to ensure a robust and computationally efficient analysis, the dataset was pre-processed to filter out any duplicate projects and to exclude projects whose state was marked as *submitted*, *canceled*, *live*, *suspended* or *started* that would have introduced additional unnecessary noise in the dataset. Irrelevant columns were also removed. This filtering step ensured that only completed and relevant projects were included in the subsequent analysis. After this optimization step, the dataset consisted of 183709 rows and 50 columns.

Following this, the dataset was enriched with additional textual features in order to facilitate the analysis. The project story, risks, and rewards sections were programmatically retrieved by web

scraping using an automated browser framework (Playwright for Python). This method simulates the normal behaviour of browsers and renders each project page and extracts specific HTML segments corresponding to the three text sections. Only publicly accessible project pages were accessed. Before scraping, Kickstarter's robots.txt file was examined, personal or private information was not collected, and random delays were incorporated to minimize server overload. By optimizing the sample before performing the web scraping, unnecessary server requests were prevented and computational resources were optimized. Furthermore, if the scraping procedure encountered CAPTCHA or access control challenges, the process was paused and it retried later, they were never bypassed. Scraping was performed on 44995 projects, to ensure that the dataset is sufficiently large to support a robust analysis, however, it is important that it still remains computationally manageable. This process enabled the extraction of unstructured textual features that were not included in the WebRobots dataset and enriched the database for natural language analysis without compromising computational efficiency.

4.4. Data Pre-Processing & Feature Engineering

After obtaining all necessary features for the subsequent research, the data was cleaned and pre-processed to produce a final dataset that is suitable for analysis and model-building (García et al., 2015). This includes several filtering, transformation and validation steps.

4.4.1. Structured Data Cleaning

The structured campaign metadata extracted from the dataset required multiple cleaning and transformation steps to ensure consistency and suitability for subsequent modeling. First, to create a binary target variable for this classification task, the original campaign status variable (*status*) was converted into a numerical indicator of success. Specifically, 1 was assigned to campaigns that were marked as "successful", while 0 was assigned to campaigns that were marked as "failed".

Handling Missing Values

Missing values were inspected for each variable, and appropriate handling strategies were implemented, as proper handling of missing data is an important step in ensuring reliable statistical analysis and model performance (Little & Rubin, 2019). Features such as *profile_name*, *profile_blurb* and *is_in_post_campaign_pledging_phase* contained a large number of missing values. Since they have limited theoretical relevance and *is_in_post_campaign_pledging_phase* even represents post-outcome information, they were removed from the dataset to reduce noise and avoid introducing bias from

excessive imputation (Kuhn & Johnson, 2013). In case of categorical variables that had a small number of missing values, imputation was prioritized to preserve sample size. The missing values in *category_parent_name* were replaced with the corresponding *category_name* values, as these observations already represented high-level categories, and this allowed the preservation of category hierarchy. Similarly, missing values for location-related variables (*location_name, location_state, location_country*, and *location_type*) were imputed with the category "Unknown", which is a common practice for handling missing categorical features (Kuhn & Johnson, 2013). All numerical variables were fully populated and didn't require any imputation. Overall, the goal was to maintain informative observations while removing uninformative or structurally incomplete features or rows that could negatively impact results.

Feature Transformation and Creation

In addition, the data types of all variables were reviewed and corrected to ensure they were stored in the appropriate format, as proper variable representation is crucial for accurate predictive analysis (Kuhn & Johnson, 2013). Date variables were converted to datetime objects to allow extraction of temporal information, while categorical variables were converted to appropriate categorical representations. Several new features were created, including *campaign_duration_days* to represent the total duration of campaigns, *days_between_created_launched* to capture the time between project creation and launch. Further temporal features were created by extracting the day, hour and month of launches, as well as an *usd_pledge_per_backer* column for diagnostic purposes. Finally, the campaign goal was converted to U.S. dollars using *static_usd_rate* variable provided in the dataset to ensure consistency and comparability. This resulted in the standardized variable *goal_converted_USD*, which was used in all subsequent analyses.

4.4.2. Text Cleaning

The textual features that were directly scraped from the Kickstarter page were stored in raw HTML format (e.g., *rewards_html*, *risks_html*, and *story_html*). These contained many markups and elements that were not directly relevant for the analysis, therefore, a systematic HTML parsing and cleaning pipeline was implemented using the Python library BeautifulSoup to extract meaningful textual and structural features. The HTML cleaning pipeline serves two main purposes: 1.) structural feature extraction and 2.) textual cleaning.

First, the HTML structure was systematically leveraged to extract additional structural metadata embedded within the page layout. By identifying consistent tags and attributes, structural information

could be captured. This structural parsing approach enabled the transformation of semi-structured HTML into both clean textual data and quantifiable campaign-level metrics. After extracting relevant features, the text was normalized and flattened into plain text format by removing HTML tags, normalizing whitespaces and removing platform-generated boilerplate elements and repetitive metadata that is not part of creator-generated campaign content. The resulting cleaned text features are stored in corresponding columns: *story_html_text, risks_html_text* and *rewards_html_text*. Cleaning the text revealed that certain campaign descriptions only consist of images without any structured textual content. Although many of these images may contain embedded text, they require further optical character recognition (OCR) processing that is not within the scope of this study (Smith, 2007). As this research focuses on structured textual features in combination with the main title images, embedded images in the story section are not used for visual analysis. As a result, campaigns lacking textual descriptions wouldn't contribute any meaningful linguistic information and could distort the text-based modeling process. For this reason, observations with missing textual content in the main description field were removed from the dataset (Bird et al., 2009). Rows with missing values in *rewards_html_text* were also removed, since the reward structure represents a central element of crowdfunding campaigns and typically contains important information about incentive levels and perceived value for backers. Missing values in risks, however, were retained, because although the "Risks and Challenges" section is recommended to be filled by the platform, it is not strictly required and many creators leave it empty. Removing these observations would unnecessarily reduce the sample size and potentially introduce bias.

Language Filtering

Following the text cleaning process, the dataset was filtered to retain only English projects to ensure linguistic consistency across the entire analysis. First, a character-based heuristic was applied to compute the ratio of Latin characters in the text that allowed the exclusion of campaigns dominated by non-Latin characters. In order to maintain a high signal-to-noise ratio, a strict threshold of 0.90 for the proportion of Latin characters was applied, prioritizing data purity over data quantity (García et al., 2015). In addition, language recognition was conducted by using the spaCy library with a language detection component that requires a minimum confidence threshold for classifying as English (blurb ≥ 0.90, story ≥ 0.80) (Honnibal et al., 2020). Both the project blurb and the first 2000 characters of the project description were evaluated and projects that did not meet either criterion were marked as non-English and removed from the dataset. This language filtering process identified 3874 projects for exclusion. Diagnostic checks revealed that the vast majority of these were most

likely not in English (e.g., Spanish n=920, French n=461). A small number of projects (81, 2.1%) included a mix of English and non-Latin characters (e.g., bilingual English-Japanese descriptions). These bilingual projects have also been removed to ensure that all retained campaigns contain exclusively English text.

4.4.3. Outlier Analysis

Finally, an outlier analysis was also performed to ensure that all campaigns included in the dataset reflect real and serious attempts. According to Hawkins (1980, p. 1) an outlier is "an observation that deviates so much from other observations as to arouse suspicion that it was generated by a different mechanism". In case of crowdfunding campaigns, extreme or unusual values may correspond to extremely high or low funding goals or unrealistic campaign parameters, for example, such as a minimum amount of text. These observations can skew statistical patterns and negatively impact the performance and robustness of predictive models (Aggarwal, 2017). Therefore, an outlier analysis was performed to ensure that all unserious campaigns are removed and the final dataset reflects typical crowdfunding patterns. Outliers were examined in key numerical variables, particularly *goal_converted_USD* combined with the length of text included on the campaign pages or the percent funded. Crowdfunding data naturally shows a strong right-skewed distribution, as most campaigns have modest funding goals, while only a few aim to raise extremely large values (Mollick, 2014). In this case the campaign goal variable, *goal_converted_USD,* also exhibits a very heavily right-skewed distribution with a median of $5000 and a 99.9th percentile of approximately $3.9M, as visible in Table 2.

Table 2. Summary statistics of project funding goals

Count	40,471
Mean	39,064.94
Std	999,282.59
Min	0.65
25%	1,500
50%	5,000
75%	13,859.62
99.9%	3,953,085.58
Max	100,000,000

Projects with a goal of $10,000,000 or higher represented only a small portion of the dataset (~0.04%) and were considered extreme anomalies. These values are far above the platform's usual norms and can disproportionately influence model training even after logarithmic transformation. Therefore,

campaigns with a target value of $10,000,000 or higher were removed from the dataset. Some campaigns also had unusually high values of goals with minimal engagement and textual descriptions. Specifically, campaigns that had goals equal or greater than $1,000,000, story descriptions shorter than 1,000 characters, and funding levels below 1% were considered to be structurally incomplete or low effort and therefore removed. Additionally, as textual features play a key role in this research, projects that had fewer than 500 characters in all their textual features combined were excluded as well, since they do not provide sufficient linguistic information for reliable modeling. Crowdfunding campaigns are naturally variable, therefore, most of the campaigns with extreme values were retained (Mollick, 2014; Kuppuswamy & Bayus, 2018). Their removal could distort the dataset by removing valuable and meaningful campaigns and thereby reduce the validity of predictions (Kuhn & Johnson, 2013). Instead, appropriate transformations were applied to reduce skewness and mitigate the impact of extreme values. Particularly, the variable goal was transformed on a logarithmic scale ($\log(1 + x)$) that compresses large values while also preserves the relative order of campaigns (Box & Cox, 1964). This approach ensures a better numeric stability and model performance and also preserves the integrity and representativeness of the dataset (Osborne, 2002).

4.4.4. Information Leakage Removal

To ensure that the predictions are truly based on pre-launch dynamics, it was essential to remove any features or text that could potentially reveal any information about the outcome of a campaign. Retaining such information would lead to data or information leakage, which is "essentially the introduction of information about the target of a data mining problem, which should not be legitimately available to mine from" (Kaufman et al., 2011, p. 556). This means that the model would gain access to future information or signals related to campaign outcomes, artificially inflating performance metrics and compromising generalizability (Kaufman et al., 2011).

First of all, since the dataset was scraped after the campaigns had already ended, it contained many features that are closely connected to campaign outcome or directly encode it. These columns include backer counts, pledge amounts and platform-specific indicators of success or visibility. These were essentially removed to prevent the models from gaining access to any indicators of success or failure. Additionally, irrelevant, noisy, and "near-zero variance predictors" were also dropped to improve the robustness of the model (Kuhn & Johnson, 2013). Beyond structured features, campaign creators are able to edit their textual campaign descriptions after launch to include any updates on campaign progress, therefore, textual features could contain words or phrases that also reveal campaign

outcomes. Examples include explicit mention of terms such as "*successfully funded*", "st*retch goal unlocked*" or phrases such as "*we reached our goal*". Retaining these may allow the model to trivially infer the target variable, therefore, they were also systematically identified and removed from the textual data before model training.

4.4.5. Dataset Splitting

The dataset was divided into training, validation, and test sets using a stratified sampling strategy based on the target variable to ensure consistent evaluation across all experimental settings. Stratification is used to ensure that successful and unsuccessful campaigns are proportionally represented in all subsets (Kohavi, 1995). The final split consisted of 70% training data, 15% validation data, and 15% test data. The training set was used to fit the models and perform all preprocessing steps, the validation set was used to guide the fine-tuning of hyperparameters and early stopping, and the test set was used for the final performance evaluation, following standard practices in machine learning (Hastie et al., 2009).

4.4.6. Metadata Feature Processing

The structured campaign metadata was processed by applying encoding and scaling techniques. First, categorical variables, particularly campaign category, country, currency, and location type, were one-hot encoded to ensure comparability of the models and to avoid introducing model-specific biases (Kuhn & Johnson, 2013). Sub-categories were excluded from the dataset because they consisted of 161 unique category names that would have introduced an unnecessarily high sparsity and would have blown the dimensionality of the feature space. Formally, for a categorical variable X with k possible categories $\{c_1, c_2, \dots, c_k\}$, the transformation maps each observation x_i to a binary vector $\mathbf{v_i} \in \{0,1\}^k$ where

$$v_{ij} = \begin{cases} 1 & \text{if } x_i = c_j \\ 0 & \text{otherwise} \end{cases}$$

This transformation ensured that the resulting features are comparable across models and they do not introduce ordinal relationships that could distort distance-based or linear models (Kuhn & Johnson, 2013). Although tree-based models are capable of handling label-encoded features inherently, a unified modeling strategy was applied to ensure consistency and interpretability.

For numerical features, scaling was used to normalize the values and minimize the impact of different units of measurement or ranges. Let x_i denote the value of a numerical feature for observation i.

Standard scaling transforms the data according to *(Bishop, 1995):*

$$x_i^{scaled} = \frac{x_i - \mu}{\sigma}$$

Where μ is the mean and σ is the standard deviation of the feature across the training set. This results in features with zero mean and unit variance and improves convergence of gradient-based algorithms and ensures that no feature dominates models due to its scale (Bishop, 1995). The parameters for standardization were learned only on the training data and then applied to the validation and test sets to prevent information leakage (Hastie et al., 2009). Only continuous variables were standardized, binary variables (e.g. presence of campaign video or reward images) and one-hot encoded features were not scaled.

4.4.7. Text Feature Processing

Textual campaign information was extracted from several text fields, including the campaign title, a short description (blurb), the campaign story, the section on risks, and the description of rewards. For both classical machine learning models and deep learning methods, a multi-step preprocessing pipeline was performed to ensure high-quality inputs. First, a natural language processing (NLP) pipeline was developed, which included converting text to lowercase and removing URLs, any remaining HTML tags, mentions, hashtags, punctuation and standalone numbers. The text was then tokenized, stopwords were removed, and each token was lemmatized according to its part-of-speech to preserve semantic meaning (Bird et al., 2009; Manning et al., 2008). This resulted in representations that are suitable for feature extraction methods such as TF-IDF and Bag-of-Words (BoW) (Salton & Buckley, 1988). For the transformer-based models like BERT, a more simple preprocessing pipeline was used, as these models only require minimal text cleaning (Devlin et al., 2018). This process is primarily for removing any residual HTML tags, normalizing whitespace and maintaining the overall text structure.

Two classical natural language processing approaches were used to transform text into numerical features. The Bag-of-Words (BoW) representation transforms text into a sparse matrix of token frequencies (Harris, 1954; Salton & Buckley, 1988). For a vocabulary of size $\boldsymbol{V}$, each document d is represented as a vector $\boldsymbol{x}_d \in \mathbb{R}^V$ where each element

$$x_{d,i} = count\ of\ term\ i\ in\ document\ d$$

The model was limited to the 5000 most frequent terms to reduce dimensionality and focus on the most informative tokens, a standard practice to mitigate the "curse of dimensionality" (Manning et al., 2008). It included both unigrams and bigrams to be able to capture both individual word occurrences and short phrase patterns. In addition, Term Frequency-Inverse Document Frequency (TF-IDF) representations were generated to emphasize terms that are distinctive for a given document (Salton & Buckley, 1988). For each term t in document d, the TF-IDF weight is defined as:

$$TF - IDF_{t,d} = \left(1 + log\left(tf_{t,d}\right)\right) \times log\left(\frac{N}{df_t}\right)$$

Where $tf_{t,d}$ is the frequency of term t in document d, df_t is the number of documents containing t, and N is the total number of documents (Sparck Jones, 1972). TF-IDF reduces the weight of frequently occurring words and increases the weight of words which are more informative in each document (Salton & Buckley, 1988). The vectorizer was limited to the 8000 most informative features and contained both unigrams and bigrams. To reduce noise, terms that appeared in fewer than five documents or in more than 90% of the documents were excluded, and sublinear term frequency scaling was also used (Manning et al., 2008). Common English stopwords have also been removed.

Sentence-BERT (all-MiniLM-L6-v2) was also used to find deeper semantic relationships within campaign descriptions. It converts the text of each campaign into a dense numerical vector that reflects the semantic meaning of that text (Reimers & Gurevych, 2019). Initially, pre-trained Sentence-BERT embeddings were used without any further training or fine-tuning. These embeddings already provide a meaningful and efficient representation of the text and do not require any computationally heavy training (Wang et al., 2020). However, to further improve the quality of these representations, a custom, fine-tuned transformer model was also trained specifically for this crowdfunding prediction task. This model used a pre-trained transformer encoder with partial fine-tuning of the final transformer layers, a projection head consisting of fully connected layers and a classification layer that predicts campaign success. During training, the encoder parameters were mostly frozen, while the final transformer layers and the projection head were fine-tuned to adapt the language model to crowdfunding-specific patterns. The model was trained using cross-entropy loss with label smoothing (Szegedy et al., 2016), the AdamW optimizer (Loshchilov & Hutter, 2017), and early stopping based on validation performance. Gradient clipping was used to stabilize the training by preventing the exploding gradient problem (Pascanu et al., 2013). After training, the model was used to extract fine-tuned text embeddings.

Two different strategies were used to construct the textual representation. First, all textual fields were combined into a single document for each campaign, and second, a structured text representation was created, where each textual field was processed separately. This preserves semantic differences between different campaign sections and allows the models to capture section-specific signals. In both cases, all three vectorization techniques, including BoW, TF-IDF and BERT embeddings, were applied.

4.4.8. Image Feature Processing

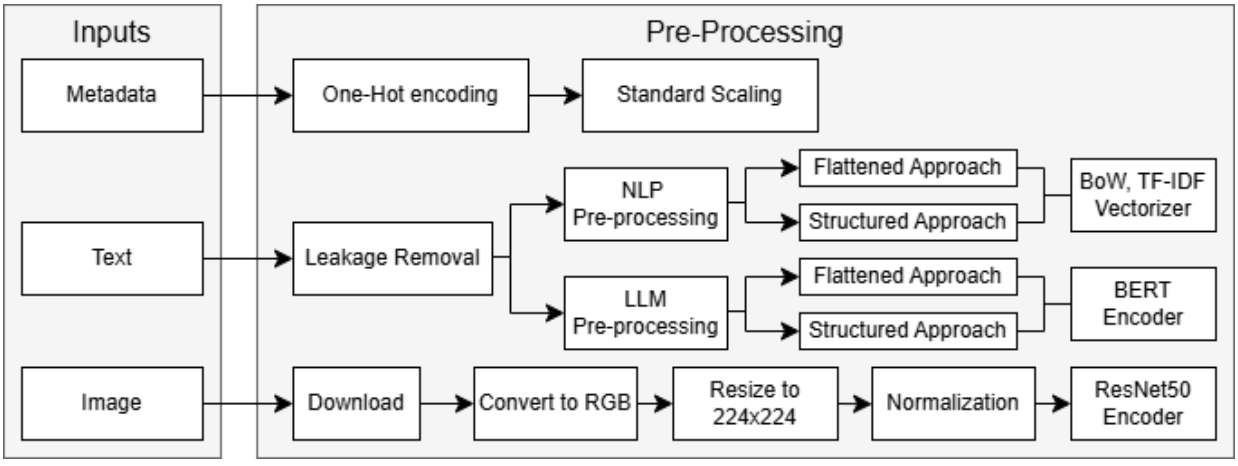
A specific process was used to generate numerical features from the images. Each image was first downloaded and resized to 224×224 pixels to meet input specifications of popular convolutional neural networks. Then features were extracted with a pre-trained ResNet50 convolutional neural network (He et al., 2016), where the classification layers were removed to obtain image embeddings. Each input image is denoted as a 3D tensor $I_i \in \mathbb{R}^{224\times224\times3}$. The ResNet50 encoder f_{ResNet} maps each image into a dense feature vector:

$$\mathbf{v}_i = f_{ResNet}(I_i), \quad \mathbf{v}_i \in \mathbb{R}^{2048},$$

where $\boldsymbol{v}_i$ encodes high-level visual patterns and semantic content of the image. Before feeding the images into the network, the pixel values were pre-processed according to the ResNet50 specifications, including normalization to fit into the distribution of the ImageNet dataset on which the model was trained (Deng et al., 2009). The obtained embeddings are a compact numerical representation of the campaign images and allow for incorporating visual information together with textual and structured features into predictive models (Sharif Razavian et al., 2014).

For a clear overview, the entire pre-processing workflow is summarized in Figure 4.

Figure 4. Multimodal data pre-processing and feature extraction framework



4.5. Final Dataset

The resulting dataset consists of structured metadata, textual inputs and visual information that provide a comprehensive view of each campaign. It has 40301 rows and 51 columns and includes campaigns from 2009 to 2025. The metadata features describe important attributes of campaign setup, timing, rewards and media usage. The textual features capture multiple campaign segments that are processed differently for traditional NLP methods and LLM inputs. The main campaign picture represents the visual information and the target variable is a binary indicator of campaign success. Table 3 provides a comprehensive overview of all features in the final dataset, their types and descriptions.

Table 3. Summary table of final dataset

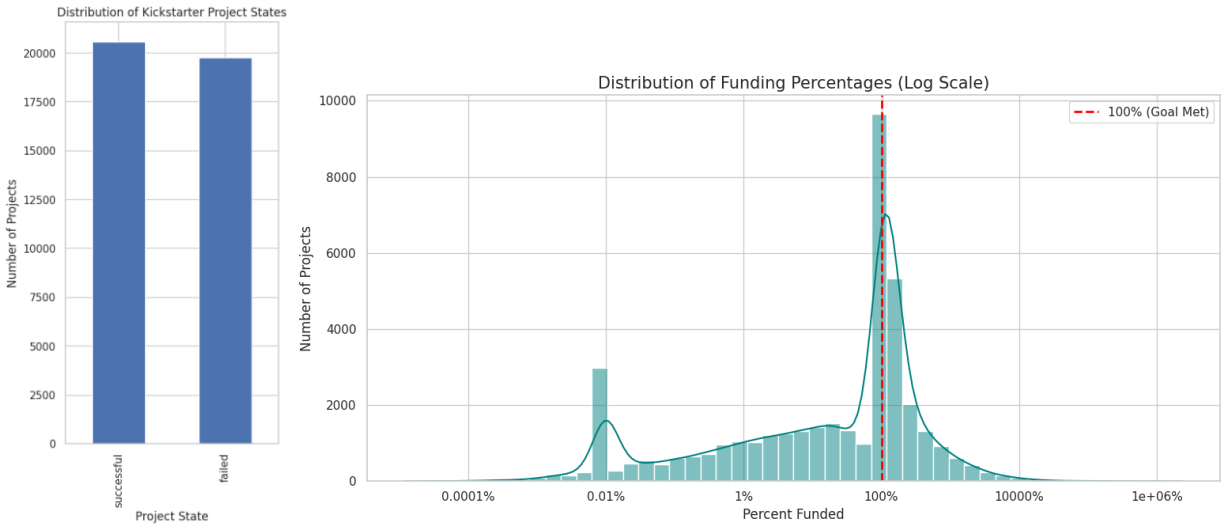
Modality		Feature name	Feature type	Description
Metadata		category_parent_name	Categorical	High-level category
		country	Categorical	Country in which the campaign was registered
		creator_has_admin_message_badge	Binary	Whether the creator has an official/admin badge
		creator_ppo_has_action	Binary	Whether the creator has prior platform-related actions
		creator_backing_action_count	Numerical	Number of past backing actions by the creator.
		currency	Categorical	Currency used for the campaign goal
		location_type	Categorical	Type of project location (e.g. town, county)
		prelaunch_activated	Binary	Whether a prelaunch phase was activated
		video	Binary	Whether the campaign includes a title video
		campaign_duration_days	Numerical	Total duration of the campaign in days
		days_between_created_launched	Numerical	Time gap between campaign creation and launch in days
		launched_day	Numerical	Day of the month when the campaign was launched
		launched_day_of_week	Numerical	Day of the week of campaign launch
		launched_hour	Numerical	Hour of the day when the campaign was launched
		launched_month	Numerical	Month in which the campaign was launched
		goal_converted_USD_log	Numerical	Log-transformed funding goal in USD
		num_reward_tiers	Numerical	Number of reward tiers offered in the campaign
		lowest_pledge_level_usd	Numerical	Price of the lowest available reward tier in USD
		highest_pledge_level_usd	Numerical	Price of the highest available reward tier in USD
		has_reward_images	Binary	Whether reward tiers include images
		average_number_of_words_per_reward	Numerical	Average length of reward descriptions in words
		has_restricted_shipping	Binary	Whether shipping is restricted
		rewards_char_length	Numerical	Character length of all reward descriptions
		blurb_char_length	Numerical	Character length of the campaign blurb
		story_char_length	Numerical	Character length of the campaign story
		risks_char_length	Numerical	Character length of the risks section
		total_text_length	Numerical	Total character length across all textual fields
		total_media_assets	Numerical	Total number of media elements in the campaign
		num_static_images	Numerical	Number of static images
		num_animated_content	Numerical	Number of animated media elements (e.g., GIFs, videos)
		ratio_text_to_total_assets	Numerical	Ratio of textual content length to total media assets
		is_local	Binary	Whether the project's physical location is the same as the registered country
Text	NLP	name_nlp	Text	Project name text, pre-processed for NLP tasks
		blurb_nlp	Text	Campaign blurb text, pre-processed for NLP tasks
		story_html_text_nlp	Text	Campaign story text, pre-processed for NLP tasks
		risks_html_text_nlp	Text	Risks section text, pre-processed for NLP tasks
		rewards_html_text_nlp	Text	Rewards description text, pre-processed for NLP tasks
	LLM	name_llm	Text	Project name text, pre-processed for LLM tasks
		blurb_llm	Text	Campaign blurb text, pre-processed for LLM tasks
		story_html_text_llm	Text	Campaign story text, pre-processed for LLM tasks
		risks_html_text_llm	Text	Risks section text, pre-processed for LLM tasks
		rewards_html_text_llm	Text	Rewards description text, pre-processed for LLM tasks
Image		photo_1536x864	URL	URL of the campaign's main image
Target		success	Binary	Whether the campaign was successful (1) or failed (0)

For some features (*blurb, name, rewards_html_text, risks_html_text, story_html_text, goal_converted_USD, category_name)*, the raw variables were retained in the dataset, but they were not used directly as input features for traditional machine learning or deep learning models.

4.6. Exploratory Data Analysis (EDA)

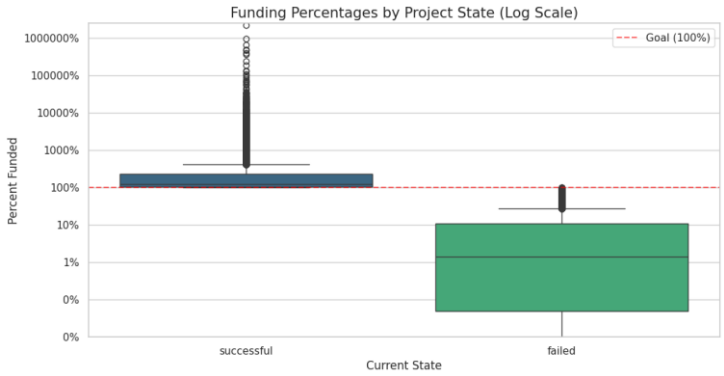
Exploratory data analysis was performed to better understand the structure and characteristics of the final dataset before model building (Tukey, 1977). In total, the final dataset comprises 40301 campaigns, of which 20556 are successful and 19745 have failed (Figure 5). This provides a quite balanced class distribution with about 51% successful projects and 49% unsuccessful projects. This is highly beneficial in machine learning tasks since class imbalance could distort the predictive performance of models by being biased towards the dominant class. The distribution of funding percentages follows a strongly right-skewed distribution and has two notable peaks (Figure 5). This confirms that the outcomes of crowdfunding campaigns are very uneven, with a large number of projects receiving little or no funding, while a smaller group of campaigns achieve very high levels of funding, often far exceeding their initial targets (Mollick, 2014).

Figure 5. Distribution of project outcomes & funding percentages



The box plots (Figure 6) provide a further visual comparison of the funding percentages achieved by successful and unsuccessful campaigns. As anticipated, successful projects consistently surpass the 100% funding mark, whereas those that are unsuccessful remain significantly below this threshold. In addition, the occurrence of extreme outliers among the successful group reinforces the earlier finding of a strongly right-skewed distribution in the data, emphasizing that a small subset of "viral" campaigns raise funds far beyond their initial targets.

Figure 6. Distribution of funding percentages by project outcome (log scale)



Success rates among Kickstarter projects differ significantly depending on the category. Figure 7 illustrates that creative fields such as comics, dance, and music tend to achieve higher rates of success, whereas areas like journalism and food see considerably lower outcomes. This distribution underscores how the type of project being funded can have a substantial impact on its likelihood of success, emphasizing the importance of category-specific dynamics in the crowdfunding landscape.

Figure 7. Success rate by category

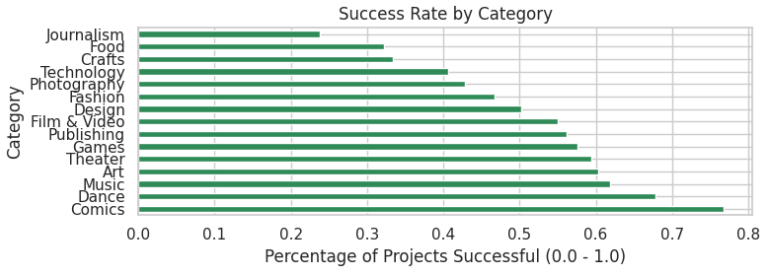
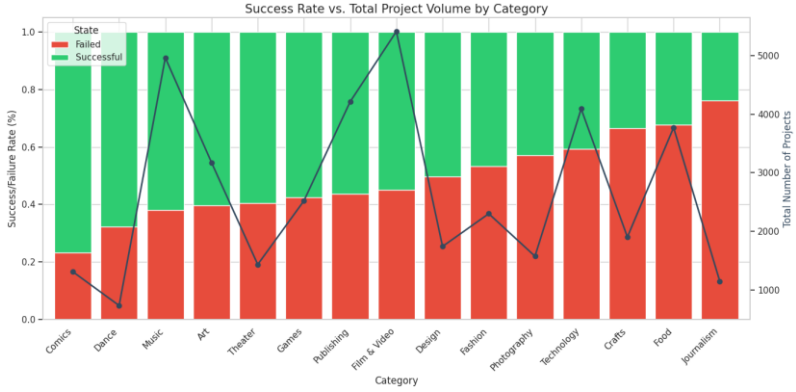


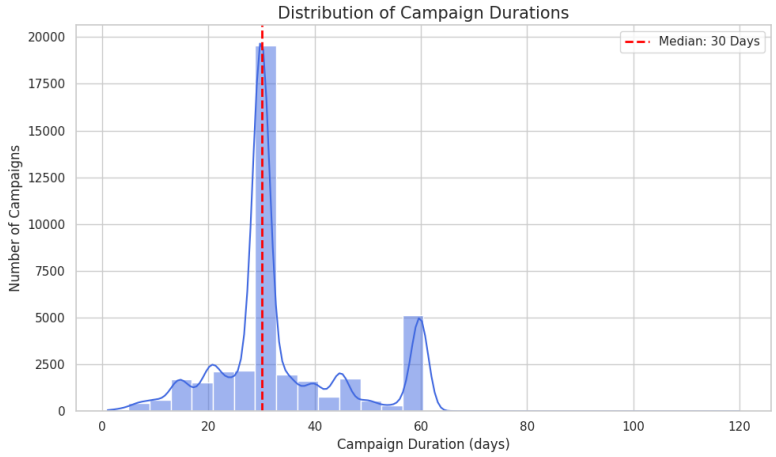
Figure 8 provides additional insight into how the number of projects in each category relates to success. It appears that smaller categories often have higher success rates, whereas those categories with a larger volume of projects tend to perform worse on average, likely reflecting stronger competition.

Figure 8. Success rate vs. total project volume by category



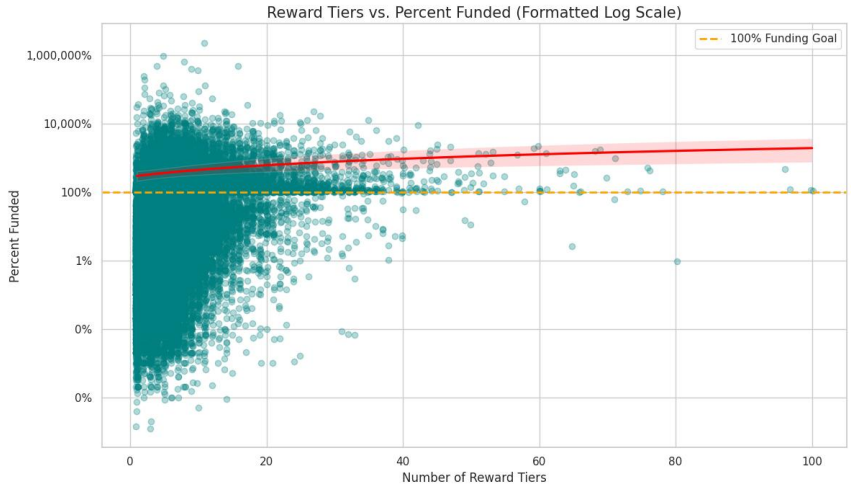
The analysis of campaign durations revealed that majority of projects run for 25-35 days with an average of 33.5 days and a median of 30 days (Figure 9). This tendency is in line with standard crowdfunding practices where projects are usually planned for a month to maintain engagement and avoid backer fatigue (Mollick, 2014; Kuppuswamy & Bayus, 2018). Only a minority of campaigns extend well beyond this timeframe.

Figure 9. Distribution of campaign durations



The regression plot visible on Figure 10 illustrates the relationship between the number of reward levels and the proportion of funding provided. Although there is significant variation in the data, the trend line shows a small positive correlation. This implies that campaigns, which present more reward options are likely to draw more supporters and raise more funds. However, the large variance in the data shows that just having a good reward structure is not sufficient to make a campaign successful.

Figure 10. Correlation between reward tiers and funding success



The median values of numeric and Boolean features show clear differences between successful and unsuccessful campaigns (Figure 11). Projects that achieve success typically demonstrate notably

greater engagement, richer content, and higher levels of backer involvement. For instance, the median total text length for successful campaigns stands at 5,890 characters, in contrast to just 3,475 for those that fail and their story and reward description lengths are also approximately twice as long as for unsuccessful projects. This suggests that campaigns are more likely to succeed if they provide more comprehensive descriptions and detailed information. Successful campaigns also tend to have more reward options (eight versus five) and set lower, more achievable funding goals. The presence of media and positive platform signals also appears to be influential, as successful projects include more media assets (a median of six compared to zero) and static images (also six versus zero), which likely helps build backer trust and increase interest. Boolean attributes such as the spotlight feature clearly differ, with successful campaigns registering a median of one while unsuccessful ones have zero, suggesting that platform endorsements may be a crucial factor. Other Boolean or engagement-related features, such as video content, creator admin badges, staff picks, and prelaunch activation, generally remain uncommon (median of zero), yet may still play a role in the success of the minority of campaigns where they are present.

Figure 11. Comparison of feature medians by project outcome

Median values for numerical features (sorted by Successful Projects):

	Successful Projects	Unsuccessful Projects
total_text_length	5,890.50	3,475.00
usd_pledged	5,184.97	85.00
story_char_length	3,257.50	1,854.00
goal_converted_USD	3,176.77	7,484.19
rewards_char_length	1,506.00	741.00
risks_char_length	474.00	400.00
highest_pledge_level_usd	420.00	250.00
ratio_text_to_total_assets	269.33	0.00
percent_funded	124.14	1.43
blurb_char_length	116.00	120.00
usd_pledge_per_backer	69.76	23.67
backers_count	68.00	3.00
campaign_duration_days	30.00	30.00
average_number_of_words_per_reward	29.80	24.25
days_between_created_launched	19.00	8.00
lowest_pledge_level_usd	9.34	10.00
num_reward_tiers	8.00	5.00
total_media_assets	6.00	0.00
num_static_images	6.00	0.00
spotlight	1.00	0.00
video	1.00	1.00
num_animated_content	0.00	0.00
creator_ppo_has_action	0.00	0.00
creator_has_admin_message_badge	0.00	0.00
creator_backing_action_count	0.00	0.00
is_disliked	0.00	0.00
disable_communication	0.00	0.00
is_starrable	0.00	0.00

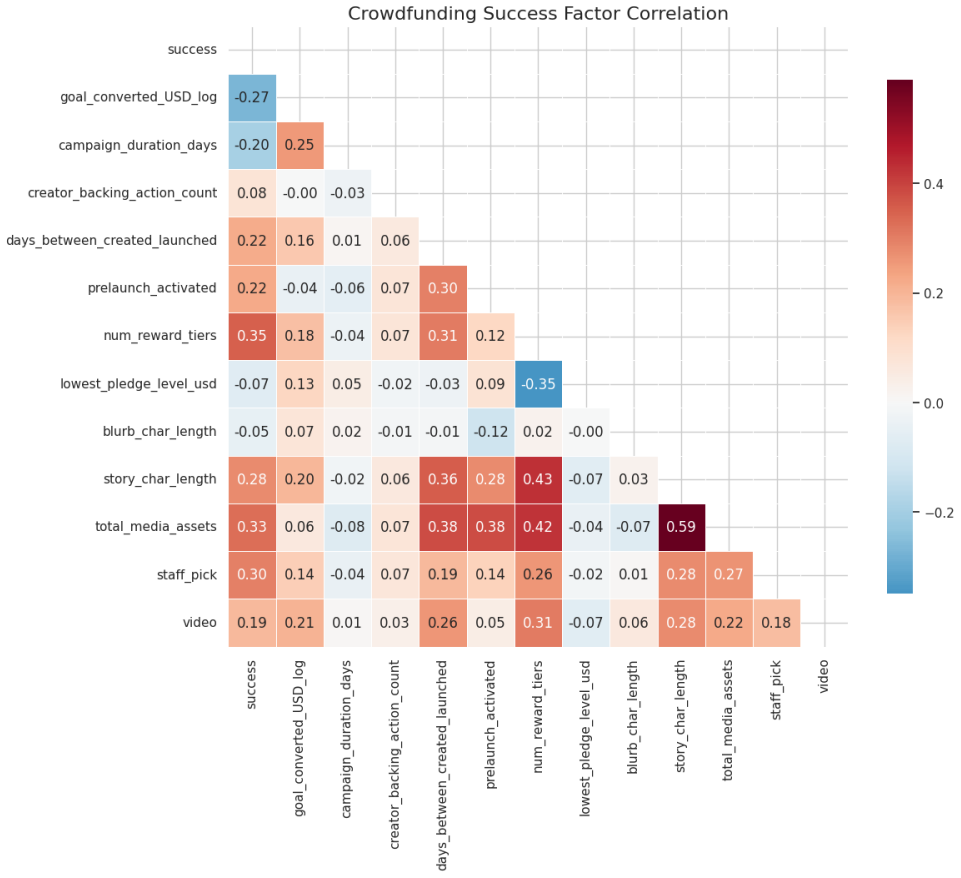
To gain deeper insight into how campaign features might be linked to project success, a Spearman correlation analysis was carried out on a selection of numeric and binary variables (see Figure 12). Spearman's Rank Correlation Coefficient was developed by Spearman in 1904 as a non-parametric

statistic used to measure the strength and direction of the monotonic relationship between two variables. The formula is presented as:

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

Where *d* is the difference between the ranks of each observation and *n* is the number of observations. Spearman's method uses the rank order of the data, unlike Pearson's correlation, which assumes a linear relationship and a normal distribution. This makes it particularly suitable for crowdfunding data, which is often characterized by highly skewed distributions and extreme outliers that would otherwise bias linear analyses (Hauke & Kossowski, 2011). Several features show moderate correlations with campaign success, including the number of reward tiers, the number of media content in the campaign, the length of the campaign story and so on. These results suggest that both campaign design factors (e.g., rewards, storytelling, media) and creator-related characteristics can influence crowdfunding outcomes. However, the majority of the correlations are still relatively low, indicating that project success is likely a function of the interaction of multiple factors, rather than one specific variable.

Figure 12. Correlation matrix of crowdfunding success factors



4.7. Model Building

4.7.1. Metadata Models

A diverse set of supervised machine learning models was trained using only structured campaign features to evaluate the predictive power of metadata on its own, pre-processed as previously described. First, Logistic Regression was implemented as a single-layer linear baseline model. Hyperparameter tuning was performed using 5-fold stratified cross-validation over the regularization strength $C \in \{0.01, 0.1, 1, 10\}$ with L2 penalty, and class_weight='balanced' was used to handle class imbalances. Feature importance directly corresponds to the learned coefficients.

The Decision Tree model was implemented as a hierarchical, single-tree classifier where each internal node is partitioned based on a single feature threshold, which is chosen to maximize the homogeneity of the resulting child nodes based on the Gini impurity criterion. The maximum tree depth [None, 5, 10, 20], minimum samples per split [2, 5, 10], and minimum samples per leaf [1, 2, 4] were tuned using 5-fold stratified cross-validation. Each leaf node provides the class probability as the ratio of the number of training samples in the positive class. The importance of the features is derived from the cumulative decrease in impurity contributed by each feature over all splits. The final output of the model is a scalar probability per sample, which is thresholded to 0.5 for binary classification.

The Random Forest model was implemented as an ensemble of decision trees trained on metadata features. Each tree in the ensemble was built on a bootstrap sample of the training data, and only a random subset of features (the square root of the total features) was considered for each split. The maximum depth of each tree [5, 10, 20], the number of estimators [100–200], and the minimum samples per leaf [2] and per split [5] were selected using 5-fold stratified cross-validation. Imbalance between classes was handled using class_weight='balanced'. Each tree outputs class probabilities, and the final probability of each sample is obtained by averaging the predictions of all trees. The importance of features in the ensemble is derived from the average decrease in impurity. The model produces a scalar probability per campaign for the likelihood of success, which can be specified as a threshold for classification.

The XGBoost model was implemented as a gradient boosting ensemble of decision trees, where each tree is trained sequentially to correct errors of the previous trees. At each boosting iteration, a tree is fitted to the gradient of the loss function relative to the current ensemble prediction. Hyperparameters including the number of estimators [100, 200, 300], maximum tree depth [3, 4, 5], learning rate [0.05–

0.1], minimum child weight [1, 3], subsample ratio [0.7, 0.9], column subsample ratio[0.7, 0.9], and gamma value [0, 0.1] were optimized using 5-fold stratified cross-validation. The final prediction for each campaign is the aggregated probability from the boosted ensemble. The feature importance is computed as the average gain of how much the feature contributes to the model's loss reduction across the ensemble of trees. LightGBM and CatBoost were also implemented as gradient-boosted ensembles of decision trees. LightGBM was trained using random search and stratified 3-fold cross-validation to select optimal hyperparameters, including the number of estimators (300–800), maximum tree depth [-1, 5, 7, 10], number of leaves (31–100), learning rate (0.01–0.1), minimum child samples, and feature subsampling ratios. Early stopping with a patience period of 50 rounds was used on the validation set to avoid overfitting and speed up training. CatBoost was trained using grid search with 5-fold stratified cross-validation to tune hyperparameters, including the number of iterations (200-300), tree depth (4-8) and learning rate (0.03-0.1). Both models accounted for class imbalance by setting the scale_pos_weight parameter as the ratio of negative to positive instances in the training set. The importance of features for both models was derived from the total gain provided by each feature across all trees.

SVMs were implemented to predict as well, including both linear and nonlinear variants. The linear SVM used a linear decision boundary and was trained with LinearSVC, which was wrapped in a CalibratedClassifierCV to enable probability estimation. First, hyperparameters were optimized using 5-fold stratified cross-validation, where the regularization parameter C was specifically optimized in the range [0.01, 0.1, 1, 10]. Once the best performing model was selected, the probability estimate was calibrated using CalibratedClassifierCV and sigmoid calibration (Platt scaling) to make it possible for the model to output well-calibrated class probabilities. These probabilistic outputs can be used as classification thresholds, and the model also provides insights on feature importance through the learned coefficients. The nonlinear SVM employed a radial basis function (RBF) kernel to account for complex, nonlinear relationships and interactions among features. Its hyperparameters, including the regularization parameter C [0.1, 1, 10] and the RBF kernel parameter γ ['scale', 0.01, 0.1], were optimized using 5-fold stratified cross-validation. The model is able to capture complex decision boundaries that linear models cannot capture. Both linear and nonlinear SVMs were robust in performance with balanced class weights to account for class imbalance.

Finally, KNN was also implemented as a non-parametric, instance-based classifier. The model calculates distances between campaigns using Euclidean or Manhattan metrics, for the number of

neighbours k ranging 3 to 11. Both uniform weighting (all neighbours contribute equally) and distance weighting (closer neighbours contribute more) were tested. Hyperparameters were tuned using 5-fold stratified cross-validation and ROC-AUC as evaluation metric. KNN does not provide feature importance as tree based or linear models, but it can capture the local patterns and the nonlinear relationships between the metadata features.

4.7.2. Metadata + Text Models

To leverage the rich semantic information in campaign texts together with metadata, two different strategies, 1.) flattened or unstructured approach and 2.) structured approach, were used to combine these features into unified representations suitable for both classical machine learning and deep learning models. In the flattened approach, text vectors from a single document were concatenated with the standardized metadata to create combined input matrices. Concatenation means joining different feature vectors end-to-end into a single continuous vector representation (Goodfellow et al., 2016). This provides a simple joint representation where all text and metadata features are treated together. In the structured approach, each text field was vectorized separately, preserving the section-specific semantics, and then these section vectors were stacked horizontally together with the single metadata vector to create structured combined matrices. This dual approach enables models to learn shared patterns from textual and structured features while preserving differences at the section level where relevant. Additionally, all text representations are preserved without concatenating with metadata. This allows deep learning models to be trained with separate branches for metadata and text inputs, allowing the model to learn joint patterns while maintaining modular input streams for structured and textual features.

The combined predictive power of the concatenated metadata and text feature representations was also assessed with similar classical machine learning models. The underlying model architectures were consistent with the metadata-only setup, with minor adjustments to accommodate the significantly increased dimensionality of the input space. Logistic Regression and Linear SVM were reimplemented with the same model configuration. To ensure convergence within the expanded feature space, the maximum iteration count was increased to 3000 and 10,000. Since LinearSVC does not produce probability estimates, probability calibration was applied again using sigmoid calibration (Platt scaling) to generate class probabilities (Platt, 1999). XGBoost was also re-implemented, albeit with a modified training architecture compared to the metadata-only configuration. Rather than employing exhaustive grid search, randomized search was utilized, incorporating 5-fold stratified

cross-validation to efficiently investigate the hyperparameter space. This included the number of estimators [300, 500], maximum tree depth [4, 5], learning rate [0.01, 0.05, 0.1], gamma value [0, 0.1, 0.2], and minimum child weight [1, 3, 5]. After hyperparameter selection, the final model was trained with early stopping on validation performance to prevent overfitting and improve computational efficiency.

Furthermore, six neural network architectures were developed to predict campaign outcomes using structured metadata and textual information (Table 4). Each model takes campaign metadata and the previously fine-tuned BERT embeddings as input. The architectures differ primarily in how they encode each modality and how they combine these representations to produce the final prediction.

Table 4. Architectural configurations for metadata + text models

Model	Metadata Branch	Text Branch	Gating / Residual	Fusion	Head
Simple Concatenation	Dense 128 → 32, ReLU, BN, Dropout 0.4/0.3	Dense 256 → 64, ReLU, BN, Dropout 0.4/0.3	-	Concatenate [meta, text] → Dense 64, ReLU, BN, Dropout 0.4	Dense 1 (sigmoid)
Deep MLP Fusion	Dense 128 → 64, Swish, BN, Dropout 0.4	Dense 256 → 128, Swish, BN, Dropout 0.4	-	Concatenate → Dense 256, swish, BN, Dropout 0.5 → Dense 64, Swish, BN	Dense 1 (sigmoid)
Gated Fusion	Dense 128 → 64, Swish, BN, Dropout 0.4	Dense 128 → 64, Swish, BN, Dropout 0.4	Scalar gating (sigmoid) based on joint features (meta + text)	Multiply text × gate + meta × (1-gate) → Concatenate	Dense 128 → 64, Swish, BN, Dropout 0.4 → Dense 1 (sigmoid)
Cross-Interaction	Dense 128 → 32, ReLU, BN, Dropout 0.35/0.25	Dense 256 → 128, ReLU, BN, Dropout 0.35/0.25	-	Concatenate → Dense 128, ReLU, BN, Dropout 0.3 → Dense 64, ReLU, BN, Dropout 0.25	Dense 1 (sigmoid)
Multi-Branch	Dense 128 → 64, Swish, BN, Dropout 0.4	5 branches (one per section) Dense 128 → 64, Swish, BN, Dropout 0.3	-	Concatenate [meta + all text branches] → Dense 256, Swish, BN, Dropout 0.5 → Dense 64, Swish, BN	Dense 1 (sigmoid)
Residual Multi-Branch	Dense 128 → 64, Swish, BN, Dropout 0.4	5 branches Dense 128 → 64, Swish, BN, Dropout 0.3	Residual metadata bypass (meta features re-injected after fusion)	Concatenate [meta + all text branches] → Dense 256, Swish, BN, Dropout 0.5 → Concatenate residual → Dense 64, Swish, BN	Dense 1 (sigmoid)

A Simple Concatenation Fusion model serves as a baseline. The metadata and textual features are encoded separately, and then combined to form a single representation, which is then passed to a final dense layer to make predictions. This method offers equal treatment to both modalities, so it is used as a baseline for more advanced fusion techniques. A Deep Fusion MLP enhances this by increasing the fusion's depth. Although the metadata and text branches remain similar to the Simple Concatenation model, the architecture incorporates the Swich activation function to facilitate improved gradient propagation within deeper layers. Furthermore, the fused representations are directed through a "diamond-shaped" MLP head, thereby enabling the network to learn more intricate, nonlinear interactions between metadata and text. The Gated Fusion Network, conversely, introduces learnable gates to dynamically adjust the contribution of each modality. The outputs from

each branch are scaled by a learned gating scalar, which highlights the most predictive input for each instance, thereby enabling the network to modulate the impact of metadata and text on a sample-by-sample basis. Following that, a Cross-Interaction Network takes a more refined approach to fusion by explicitly modeling the interactions between individual metadata and text features through additional dense layers. By directly processing the interactions between metadata and text features, this model captures patterns that could potentially be lost during simple concatenation. Next a Multi-Branch Sectional Network uses the decomposed textual features as the textual input. Each text section has its own dedicated feedforward encoder, allowing the model to extract section-specific features independently. This preserves subtle semantic differences across different parts of the text, differences that might be lost if the text were treated as a single unit. Finally, a Residual Multi-Branch Network builds on the multi-branch approach by incorporating a residual connection to metadata features. The metadata is incorporated both in the initial fusion stage and again in a residual bypass just before the final dense layers. This ensures that essential metadata information remains influential in the network. This architecture preserves important structured information, while enabling rich interactions with text features.

All models were trained using the Adam optimizer, and learning rates were dynamically adjusted using a ReduceLROnPlateau scheduler. Dropout and L2 weight decay (1e-5 for simpler models and 2e-5 for deeper networks) were used to regularize the models, as well as early stopping of the monitored validation AUC to prevent overfitting while restoring the best weights. The objective function was binary cross entropy loss with label smoothing. The batch sizes were modified in accordance with the model's complexity, with values ranging from 32 for the most intricate residual networks to 64 for less complex architectures. Such training decisions allowed for consistent model comparisons as well as allowing each architecture to take advantage of the full structured and unstructured inputs.

4.7.3. Metadata + Text + Image Models

To exploit the complementary strengths of all three data modalities, a series of multimodal neural network architectures have been developed. They integrate structured metadata, textual information, and image features. Each model follows a modular architecture where modality-specific branches first learn compact representations independently, which are then combined through a fusion mechanism and fed to a prediction head for final classification (Table 5).

The metadata branch processes structured campaign features using fully connected layers with nonlinear activations, batch normalization, and dropout. It handles textual modality using the sectional approach, based on fine-tuned BERT embeddings. These sectional representations are then concatenated and optionally compressed into a single textual embedding. Image features are similarly transformed into lower-dimensional representations through dense layers with regularization.

Table 5. Architectural configurations for multimodal models

	Metadata Branch	Image Branch	Text Branch	Gating/Residual	Fusion	Head
1.	Dense 128 → 64, Swish, BN, Dropout 0.4	Dense 128 → 64, Swish, BN, Dropout 0.3	Dense 128 → 64, Swish, BN, Dropout 0.3 per section	Residual bypass (meta only)	Concatenate [meta, img, text] + deep interaction (Dense 256, Swish, BN, Dropout 0.5)	Residual Concatenate (meta) → Dense 64, Swish, BN → Dense 1 → Sigmoid
2.	Dense 128 → 64, ReLU, BN, Dropout 0.4 + gating	Dense 256 → 64, ReLU, BN, Dropout 0.5 + gating	Dense 256 → 64, ReLU, BN, Dropout 0.4 per section → Concatenate → Dense 128 + gating	Multiplicative gating per **modality** (sigmoid) applied to meta (64), text (128), image (64)	Concatenate [gated meta, gated text, gated image]	Dense 256, ReLU, BN, Dropout 0.5 → Dense 128, ReLU, Dropout 0.4 → Dense 1 → Sigmoid
3.	Dense 128 → 64, Swish, BN, Dropout 0.4	Dense 256 → 128, Swish, BN, Dropout 0.4	Dense 256 → 64 Swish, BN, Dropout 0.4 per section → concatenate → Dense 128	Multiplicative gating per **feature** (sigmoid) applied to meta (64), text (128), image (128)	Concatenate [gated meta, gated text, gated image]	Dense 256, Swish, BN, Dropout 0.5 → Dense 64, Swish, BN → Dense 1 → Sigmoid

In the simplest approach, the so-called Tri-Modal Residual Multi-Branch, the learned embeddings from the metadata, text, and image branches are concatenated and processed through additional dense layers to capture interactions between modalities. An additional residual connection from the metadata branch is also incorporated to preserve important structured information and stabilize learning. This connection directly feeds the learned metadata representation to later stages of the network, bypassing part of the fusion and transformation process. As a result, the model can still see the original structured features even after using complex, intermodal interactions. Thus, it decreases the chance that these highly informative signals will be contaminated or diluted. The residual path also helps the gradient flow in backpropagation, solving vanishing gradient problems and providing more stable training. This is especially helpful in multimodal settings, where deeper fusion layers might otherwise hinder the learning of modality-specific contributions.

To improve flexibility, in the next model a gated fusion architecture was introduced that applies multiplicative gating mechanisms to each modality before fusion. In this Tri-Modal Gated Fusion model, the embedding for each modality (metadata, text, and image) is multiplied element-wise by its own learned gate, which is generated using a sigmoid activation function. This setup allows the network to dynamically adjust the effect of each modality to fit the particularities of each input. This technique is a kind of soft attention mechanism over modalities, which allows the model to suppress the less informative inputs and focus on more predictive inputs for a given instance.

The final model extends this idea to the level of features, which is called Feature-Level Gated Fusion. Instead of having a single gate for the whole modality, this approach gates each feature dimension separately. This enables the model to operate on individual features independently, selectively enhancing or downplaying them. This kind of fine-grained gating is particularly useful for high dimensional data such as text embeddings, where some features may be much more important for prediction than others.

Integrated representation is fed into a shared prediction head comprising fully connected layers with nonlinear activation functions, batch normalization, and dropout for regularization. These layers further investigate and model the interactions between modalities. The last layer of the output uses the sigmoid activation to output the probability of success of the campaign. Consistent across all models, training is performed using binary cross-entropy loss with label smoothing, and optimization is carried out with the Adam optimizer. Early stopping based on validation AUC and adaptive learning rate reduction are used to ensure stable training and avoid overfitting.

To further explore the predictive potential of multimodal campaign information, Google's Gemini-2.5 Flash model was used in a multimodal setting using few-shot prompting. A small, random subset of 555 campaigns was selected from the test set to keep API usage manageable and stay within rate limits. For each campaign, the majority of structured features (e.g., category, country, funding goal, duration, number of reward tiers) and the full campaign story and the main campaign image were included. The campaign image was downloaded and processed using the Python Imaging Library (PIL) to ensure compatibility with the multimodal model. To guide the model, a few labeled examples (one successful and one unsuccessful campaign) were included in the prompt, each of which contained both textual and visual information. Then the target campaign was added and the model was asked to predict the outcome (1 = successful, 0 = failed) and give a short textual explanation. The model was configured with the google-generativeai Python API and generated predictions for each campaign. The raw responses from the model were stored and analyzed to extract the predicted binary outcome (llm_pred). The predictions were compared to the true success labels to calculate accuracy, and the errors were analyzed to identify cases where the model overestimated the results of the campaigns (false positives).

4.7.4. Hyperparameter Tuning

In order to enhance model performance and promote strong generalization, hyperparameter tuning was carried out for all traditional machine learning models. Hyperparameters are configuration

settings defined before the training process begins, not derived from the training data itself, in contrast to model parameters, which are directly learned from the data (Goodfellow et al., 2016). For most algorithms a systematic search strategy was employed with stratified cross-validation to obtain the best hyperparameter combinations. For models with a relatively limited and well-structured set of hyperparameters, such as Logistic Regression, Decision Trees, Random Forests, KNN, and SVMs, grid search was performed with 5-fold stratified cross-validation. Grid search is a systematic hyperparameter optimization technique. It requires training and validating the model for each possible combination of the specified hyperparameter values (Bergstra & Bengio, 2012). Cross-validation is a resampling technique that divides the training data into *k* number of splits called "folds". During each iteration it uses one fold for validation and the other k−1 folds for training (Stone, 1974). Stratified splitting ensured that each fold preserved the original class distribution, which made performance estimates reliable even with unbalanced datasets (Kohavi, 1995). The average cross-validated ROC-AUC score was used to choose the final configuration of the model. For the more complex models that have much larger and more complex hyperparameter spaces, such as XGBoost and LightGBM, a randomized search approach was used to efficiently sample a diverse set of parameter combinations. It randomly samples a fixed number of parameter configurations instead of exhaustively exploring all possible combinations (Bergstra & Bengio, 2012). This method decreases computational costs while still discovering strong configurations (Bergstra & Bengio, 2012). For these models, cross-validation was combined with early stopping based on performance on a separate validation set. Training was stopped when no further improvement was seen to avoid overfitting and speed up training time. The final hyperparameter configurations were chosen based on cross-validated performance and then tested on the test set to get an unbiased estimate of predictive performance.

Hyperparameter tuning for deep learning models focused on both the network's structure and its training process to achieve strong performance and generalization. Regarding architectures key choices included the number of dense layers, the number of units in each layer and the selection of activation functions (either ReLU or Swish), as these affect the model's ability to learn complex patterns. Fusion strategies, such as simple concatenation, gated fusion, cross-interaction, and multi-branch structures, were set for each model, but the size and number of dense layers after the fusion step were adjusted to avoid overfitting while maintaining enough flexibility. Training parameters like the learning rate, dropout rate, L2 regularization (weight decay), batch size, and label smoothing were also systematically explored. The learning rate was managed dynamically with a ReduceLROnPlateau scheduler. Dropout and weight decay were tuned to avoid overfitting. Early

stopping based on validation AUC was used to stop training when no further improvement was observed. For models with multiple text segments, the number of units in each segment-specific branch and the dimensionality of the fusion representations were also tuned. Hyperparameters were selected using stratified cross-validation and evaluation using a dedicated validation set. The final model configurations were selected based on the highest validation AUC and then evaluated on the test set.

4.8. Model Evaluation and Performance Metrics

The performance of the proposed machine learning and deep learning models is evaluated using several key metrics to get a comprehensive picture of their predictive ability. The main objective of this study is to predict the success of campaigns as a binary classification task, therefore, the evaluation strategy is based on widely used standard methodologies for binary classification problems. All evaluation metrics are consistently calculated on the same test set in all modality settings to ensure a fair comparison between models. Using multiple evaluation metrics enables a more balanced assessment of the model's performance and a substantive comparison between traditional and more advanced approaches.

Confusion Matrix

Model performance is first assessed using the confusion matrix. The confusion matrix compresses the results of predictions into four categories and forms the basis of all the following evaluation metrics. The four main components of the confusion matrix are explained as follows:

- True Positives (TP): Successfully funded campaigns correctly predicted as successful
- True Negatives (TN): Unsuccessfully funded campaigns correctly predicted as unsuccessful
- False Positives (FP): Unsuccessfully funded campaigns incorrectly predicted as successful
- False Negatives (FN): Successfully funded campaigns incorrectly predicted as unsuccessful

Accuracy

Accuracy measures the proportion of observations that were correctly classified out of the total number of sampled observations. It is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy is an obvious measure of model performance, but it can be misleading in case of imbalanced data or when some types of prediction errors are more costly than others.

Precision

$$Precision = \frac{TP}{TP + FP}$$

Precision examines how many of the campaigns predicted to be successful are actually successful. A high precision value means that there are few false positives or that the model rarely classifies failed campaigns as successful. This is particularly important when the cost of wrongly classifying a failed campaign as successful can be very high.

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Recall (also known as sensitivity or true positive rate) is the proportion of successful campaigns that the model correctly identifies. A high recall rate demonstrates that the model is effective at identifying most successful campaigns. It is important when missing successful campaigns (false negatives) is undesirable.

F1-Score

The F1-Score measures the harmonic mean of precision and recall:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is a balanced evaluation metric when both false positives and negatives are important, and it is especially useful when class distribution is uneven.

Receiver Operating Characteristic (ROC) Curve and AUC

In addition to threshold-dependent metrics, the model's discrimination ability is evaluated using the ROC (Receiver Operating Characteristic) curve. The ROC curve shows the relationship between the true positive rate (Recall) and the false positive rate for different classification thresholds. The area under the ROC curve (AUC-ROC) gives a single value between 0 and 1 that summarizes this relationship. An AUC of 0.5 means random guessing, while an AUC of 1.0 means perfect discrimination.

In this study it is used as the primary evaluation metric. AUC measures the probability that the model assigns a higher predicted score to a randomly chosen positive example than to a randomly selected negative example. Unlike accuracy, AUC does not depend on a specific decision threshold and

remains stable even when the class distribution changes. This makes it particularly useful for predicting crowdfunding success for several reasons. First, crowdfunding datasets often exhibit some degree of class imbalance, where accuracy can give misleadingly optimistic results if the model favours the majority class. Additionally, a lot of the models used in this study, like Logistic Regression and tree-based methods, produce class probabilities, not just fixed class labels. AUC measures how well these probabilities rank the outcomes, which makes it especially useful for comparing models with different decision processes and structures. The purpose of this study is more than just simple binary classification and in practice it is often more useful to distinguish campaigns with higher or lower probabilities than relying on a single cutoff point. AUC evaluates the model's ability to distinguish between successful and unsuccessful campaigns across all possible thresholds, thus offering a more comprehensive assessment of predictive quality. Therefore, AUC provides a more reliable indicator of ranking and separation ability in this binary prediction task and is chosen as a primary metric.

4.9. Evaluation of Feature Importance

This research also aims to understand which features contribute most to the success of crowdfunding campaigns. Assessing these features contributes to the interpretation of models and offers an insight into main factors influencing their predictions. In models that include importance measures, they directly show how much each feature contributed to the predictions made by the model. This contribution can come from its effect on decision boundaries or from its role in reducing the model's training loss. However, these measurements may vary depending on the model type, and they can be influenced by feature scaling, correlations or model structure. Therefore, relying solely on model-specific importance can lead to inconsistent or biased interpretations.

4.9.1. SHAP Values for Model-Agnostic Interpretation

In order to address the limitations of model-specific feature importance measures and provide a more consistent interpretation, SHAP (SHapley Additive Explanations) was implemented (Lundberg & Lee, 2017). It is based on cooperative game theory and the concept of Shapley values, that were originally introduced for the fair distribution of payments between players in a coalition (Shapley, 1953). A Shapley value is the average marginal contribution of a player to all possible coalitions (Ponce-Bobadilla et al., 2024). In machine learning tasks we treat all features as players in a game, and the prediction of the model is the final payout (Lundberg & Lee, 2017). SHAP values calculate how much each feature contributes to the prediction by averaging its impact on all possible feature

combinations (Ponce-Bobadilla et al., 2024). This provides a fair and consistent method for determining the importance of specific features.

SHAP satisfies three important theoretical properties (Lundberg & Lee, 2017):

1. **Local Accuracy (Additivity)** - The sum of all feature contributions is equal to the difference between the model's prediction and the baseline (expected value): $\sum_{i=1}^{M} \phi_i = f(x) - E[f(x)]$
2. **Missingness** - Features that are missing from the input (or have no impact on the outcome) are assigned a SHAP value of zero.
3. **Consistency (Monotonicity)** - If a model changes so that a feature's marginal contribution increases or stays the same regardless of other features, its SHAP value will not decrease.

These properties make SHAP theoretically more robust than traditional impurity-based feature importance metrics (Lundberg & Lee, 2017; Molnar, 2019). Additionally, one of the main advantages of SHAP is that it provides global and local interpretation as well (Lundberg & Lee, 2017). The global interpretation is obtained from aggregating the SHAP values of all observations. This reveals which features have the highest cumulative effect on model predictions (Lundberg & Lee, 2017). Local interpretation explains individual predictions by showing how specific features influence the prediction towards success or failure (Lundberg & Lee, 2017). This dual interpretation is particularly valuable in research on crowdfunding. For example, funding goal can be one of the most prominent factors globally, but if we inspect a campaign with a high funding goal locally, it may still receive a high predicted probability of success if it accounts for the risk with other mitigating factors. SHAP values also provide information on both the magnitude and direction of the impact of features, unlike traditional tree-based importance measures (Lundberg & Lee, 2017; Molnar, 2019). A positive SHAP value increases the predicted probability compared to the baseline (for example, it increases the predicted probability of a campaign's success), while a negative SHAP value decreases it (Lundberg & Lee, 2017). Applying SHAP values to multiple models makes it possible to compare whether key features remain consistent across different modeling approaches (Molnar, 2019). They also improve the transparency and interpretability of models and address the common criticism that machine learning models and ensemble methods are "black boxes" (Lundberg & Lee, 2017). It also facilitates deeper theoretical integration and thus bridges the gap between machine learning methodology and the theory of crowdfunding.

5. Results

This section presents the empirical performance of the models for different feature representations and configurations. The results are evaluated using standard classification metrics and allow for comparison of modeling approaches. First, to contextualize the performance of the models developed in this study, Table 6 summarizes the results of previous research on crowdfunding success prediction. These studies differ in their use of pre- and post-launch information, they use various combinations of input modalities, as well as different machine learning and deep learning methods, but they are all based on data from Kickstarter, which provides a useful basis for comparison and for interpreting the results obtained in this work. The size of the datasets varies considerably, ranging from as few as 420 observations (Doula & Chowdhury, 2025) to over 480,000 campaigns (Abuamer et al., 2025), with most studies using samples of roughly 8,000 to 190,000. Most of the datasets used are moderately imbalanced, with the rate of success usually between 36% and 55% (e.g., Cheng et al., 2019; Kaminski & Hopp, 2020; Gunduz et al., 2026), although some studies use balanced datasets (e.g., Cai et al., 2024; Wang et al., 2020).

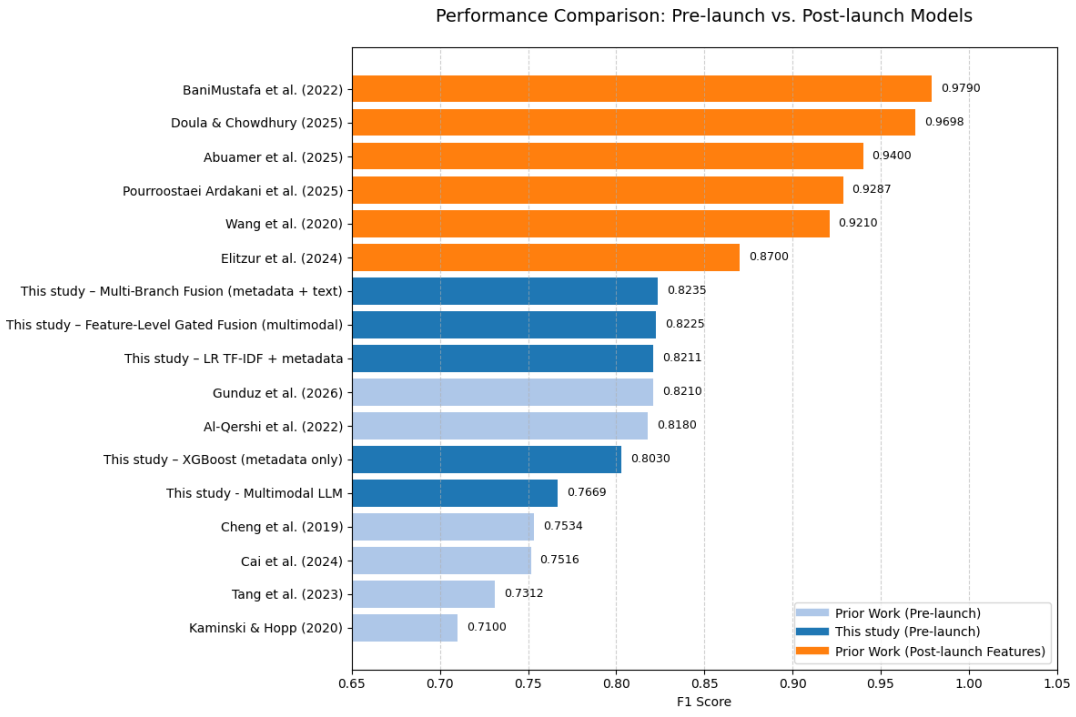
Table 6. Comparison of existing literature on crowdfunding success prediction

Paper	Method	Dataset size	Features	ROC-AUC	Accuracy	F1
Cheng et al. (2019)	Multimodal Deep Learning	34,112 55% success rate	Metadata + Text + Images Pre-launch features	**0.8326**	-	0.7534
Kaminski & Hopp (2020)	Logistic Regression	20,188 39% success rate	Text + Speech + Video Pre-launch features	-	0.71	0.71
Al-Qershi et al. (2022)	XGBoost	15,195	Metadata + Text + Speech and Visual content from video ads Pre-launch features	-	**0.818**	0.818
Tang et al. (2023)	Deep Cross-Attention Network	8868 51% success rate	Text + Video Pre-launch features?	-	0.7277	0.7312
Cai et al. (2024)	Multimodal Dynamic Graph Convolutional Network	8000 50% success rate	Text + Video Pre-launch features	0.7994	0.7369	0.7516
Gunduz et al. (2026)	CBAM + ARO + LSTM	191,724 40% success rate	Text (blurb only) Pre-launch features	-	0.778	**0.821**
Yu et al. (2018)	Multilayer Perceptron	378,611 36% success rate	Metadata Post-launch Features	0.9323	0.9320	-
Wang et al. (2020)	Multilayer Perceptron	85,233 50% success rate	Metadata Post-launch Features	0.973	0.923	0.921
BaniMustafa et al. (2022)	KNN	300,00	Metadata Post-launch Features	**0.983**	-	**0.979**
Elitzur et al. (2024)	Gradient Boosted Trees	108,223 44% success rate	Metadata Pre-launch features*	0.93	-	0.87
Zhang & Lau (2024)	CNN	38,190 73% success rate	Metadata + Text + Audio + Video Post-launch features	-	0.822	-
Abuamer et al. (2025)	Bidirectional LSTM	482,695	Metadata + Text (blurbs only) Post-launch features	0.9807	0.93	0.94
Doula & Chowdhury (2025)	Random Forest	420	Metadata Post-launch features	-	**0.9642**	0.9698
Pourroostaei Ardakani et al. (2025)	SVM	24,100	Metadata Post-launch features	-	0.9576	0.9287

*Note: While categorized as pre-launch in the original study, the inclusion of social capital metrics derived from comments received during the campaign introduces post-launch information.

The results reported in previous studies show a clear performance difference between those that rely on post-launch information and pre-launch information. Studies that consider post-launch features such as number of backers, early funding dynamics, and pledged amount consistently show better predictive performance (e.g., Wang et al., 2020; BaniMustafa et al., 2022; Abuamer et al., 2025). Accuracy values often exceed 0.93 and in some cases they even achieve near perfect classification, as shown in Figure 13. Elitzur et al. (2024) stated that their analysis was based only on pre-campaign data but reported impressive results (ROC-AUC=0.93 and F1=0.87). However, such high performance is likely to be compromised by the inclusion of social capital measures that were based on comments posted during the campaign, effectively introducing post-launch information into their model. It serves as a post-campaign signal, therefore, it will be treated as a post-launch study going forward. This is expected because these variables directly reflect campaign results or near-outcome signals, which greatly facilitates the prediction task and introduces some temporal information leakage. In contrast, models that rely solely on pre-launch information, which is the focus of this study, achieve more moderate but methodologically meaningful levels of performance (Figure 13). These models are tested in a stricter and more realistic setting, where details about the attractiveness of the campaign (e.g., backers or pledged amount) are not available at the time of prediction. Within these constraints, performance reported in the literature typically ranges from 0.71 to 0.87 F1 values for multimodal deep learning approaches (e.g., Cheng et al., 2019; Cai et al., 2024).

Figure 13. Performance comparison of prior studies and some of the proposed models in this study



In this study the performance of models trained on structured metadata alone achieves strong performance (Table 7) compared to previously published results that do not rely on post-launch metrics (Figure 13).

Table 7. Performance of metadata-only models

Model name	ROC-AUC	Training ROC-AUC	Accuracy	Precision	Recall	F1
Logistic Regression	**0.853**	**0.847**	**0.772**	**0.77**	**0.77**	**0.77**
Decision Tree	0.807	0.890	0.743	0.74	0.74	0.74
Random Forest	0.859	0.995	0.778	0.78	0.78	0.78
XGBoost	**0.876**	**0.915**	**0.792**	**0.79**	**0.79**	**0.79**
LGBM	0.874	0.911	0.794	0.79	0.79	0.79
CatBoost	0.876	0.919	0.794	0.79	0.79	0.79
Linear SVM	**0.853**	**0.847**	**0.774**	**0.76**	**0.76**	**0.76**
Non-linear SVM (RBF)	**0.865**	**0.886**	**0.779**	**0.78**	**0.78**	**0.78**
KNN	0.822	1.0	0.746	0.75	0.75	0.75

Among the evaluated approaches, gradient boosting models perform best, XGBoost and CatBoost achieved the highest ROC-AUC score (0.876), closely followed by LightGBM (0.874). These models also demonstrate the highest accuracy (around 0.792–0.794), with balanced precision, recall, and F1 scores around 0.79. Importantly, their training ROC-AUC scores (around 0.91–0.92) indicate relatively good generalization compared to other complex models. These results are among the highest for pre-launch prediction settings and outperform many previously proposed approaches, including more complex multimodal and deep learning models (Figure 13). In contrast, Random Forest and KNN show clear signs of overfitting with extremely high training ROC-AUC scores (0.995 and 1.0). The Decision Tree model also exhibits similar behaviour, with a training ROC-AUC of 0.890 compared to a validation ROC-AUC of 0.795, and its relatively low accuracy (0.732) further emphasizes the limitations of single-tree models in capturing stable patterns in the data. This big gap between training and test results suggests that they capture noise and too complex patterns that do not generalize well to unseen data. Linear models like Logistic Regression and linear SVM have slightly lower ROC-AUC scores (0.838), but their performance is consistent on both training and validation sets, showing better generalization. Boosting models offer a good balance between accuracy and generalizability, while linear models remain robust and reliable.

Introducing textual features to metadata consistently improves the predictive performance of models (Table 8). Linear models using TF-IDF representations achieve the best results, with Logistic Regression and SVM achieving ROC-AUC values of 0.904 and accuracies of 0.816–0.817, outperforming both Bag-of-Words and BERT-based features. Structured feature representations provide some minor improvements in some cases but are generally limited. Pre-trained or fine-tuned

BERT embeddings perform well with ROC-AUC scores between 0.888 and 0.891, but do not outperform TF-IDF approaches. When combining the text with metadata, the benefits are also shown by the XGBoost models, with ROC-AUC scores up to 0.900, but still not exceeding the best linear models. All methods report consistently balanced precision, recall, and F1 scores (around 0.81–0.82), reflecting reliable classification. The results show that predictions can be significantly improved by combining the textual features with the metadata. In particular, TF-IDF-based linear models offer the optimal balance of accuracy, robustness, and simplicity.

Table 8. Performance of metadata + text models

	ROC-AUC	Accuracy	Precision	Recall	F1
LR BoW + metadata	0.874	0.792	0.79	0.79	0.79
LR structured BoW + metadata	0.876	0.796	0.80	0.80	0.80
LR TF-IDF + metadata	**0.904**	**0.816**	**0.82**	**0.82**	**0.82**
LR structured TF-IDF + metadata	0.898	0.811	0.81	0.81	0.81
LR BERT + metadata	0.878	0.796	0.80	0.80	0.80
LR structured BERT + metadata	0.890	0.804	0.80	0.80	0.80
LR fine-tuned BERT + metadata	0.889	0.803	0.80	0.80	0.80
LR structured fine-tuned BERT + metadata	0.898	0.812	0.81	0.81	0.81
SVM BoW + metadata	0.877	0.794	0.79	0.79	0.79
SVM structured BoW + metadata	0.878	0.795	0.79	0.79	0.79
SVM TF-IDF + metadata	**0.904**	**0.817**	**0.82**	**0.82**	**0.82**
SVM structured TF-IDF + metadata	0.898	0.811	0.81	0.81	0.81
SVM BERT + metadata	0.878	0.794	0.79	0.79	0.79
SVM structured BERT + metadata	0.890	0.806	0.81	0.81	0.81
SVM fine-tuned BERT + metadata	0.888	0.801	0.80	0.80	0.80
SVM structured fine-tuned BERT + metadata	0.897	0.811	0.81	0.81	0.81
XGBoost BoW + metadata	0.898	0.813	0.81	0.81	0.81
XGBoost structured BoW + metadata	0.898	0.815	0.82	0.81	0.81
XGBoost TF-IDF + metadata	**0.900**	**0.815**	**0.82**	**0.81**	**0.81**
XGBoost structured TF-IDF + metadata	0.897	0.813	0.81	0.81	0.81
XGBoost BERT + metadata	0.886	0.803	0.80	0.80	0.80
XGBoost structured BERT + metadata	0.890	0.806	0.81	0.81	0.81
XGBoost fine-tuned BERT + metadata	0.891	0.814	0.81	0.81	0.81
XGBoost structured fine-tuned BERT + metadata	0.897	0.814	0.81	0.81	0.81

More advanced embeddings or structured representations provide only marginal gains and do not exceed the performance of straightforward linear models in this multimodal setting. Compared to previous studies, the results from this work are particularly compelling for predicting outcomes before launch. The best-performing configuration, LR combining metadata and TF-IDF features, achieves much better results than several prominent multimodal deep learning architectures. Specifically, this model outperforms the results of all previous pre-launch studies, including the multimodal deep learning approach of Cheng et al. (2019) (ROC-AUC 0.833) and the recent multimodal dynamic graph convolutional network (MDGCN) of Cai et al. (2024) (ROC-AUC 0.799) or the CBAM-ARO-LSTM framework of Gunduz et al. (2026) (Accuracy 77.8%, F1 0.821), despite that these models employ more complex visual, attention-based and graph-based architectures (Figure 13).

Six deep neural network architectures were also evaluated for their ability to predict campaign outcomes, using both structured metadata and textual embeddings (Table 9). The Residual Multi-Branch network had the best results, with the highest ROC-AUC (0.9002) and accuracy (0.8144), as well as strong precision, recall, and F1 score. The Multi-Branch and Gated Fusion networks also performed well with the ROC-AUC above 0.89 and F1 score above 0.82 for both. The Simple Concatenation model had the lowest scores on all metrics, but it still reached a ROC-AUC of 0.8867 and an F1 score of 0.8078. All of these models performed exceptionally well compared to previously reported results in recent years.

Table 9. Performance of different text + metadata deep learning models

	ROC-AUC	Accuracy	Precision	Recall	F1
Simple Concatenation	0.8867	0.8014	0.7975	0.8184	0.8078
Deep MLP Fusion	0.8909	0.8047	0.7841	0.8515	0.8164
Gated Fusion	0.8916	0.8076	0.7873	0.8534	0.8190
Cross-Interaction	0.8904	0.8053	0.8007	0.8233	0.8118
Multi-Branch	0.8993	0.8113	0.7875	**0.8628**	**0.8235**
Residual Multi-Branch	**0.9002**	**0.8144**	**0.8104**	0.8304	0.8203

To exploit structured metadata, textual content, and visual information from crowdfunding campaigns, three multimodal deep learning architectures were developed and evaluated: the Tri-modal Residual Multi-Branch model, the Tri-Modal Gated Fusion model, and the Feature-Level Gated Fusion model (Table 10). Each model combined campaign metadata, image embeddings, and sectional text representations to predict campaign success. The Tri-Modal Residual Multi-Branch Model achieved a test ROC-AUC of 0.8956 and a precision of 0.8091, with balanced precision (0.8063) and recall (0.8236). The residual bypass allowed the metadata features to contribute directly in addition to the fused deep representations. The Tri-Modal Gated Fusion Model showed the best overall performance with an ROC-AUC of 0.9011 and precision of 0.8154. The gating of the branches contributed to the adaptive weighting of the metadata, image and text inputs, leading to a precision of 0.8168 and recall of 0.8226. The Feature-Level Gated Fusion Model achieved an ROC-AUC of 0.8950 and a precision of 0.8086. Fine-grained feature-level gating improved recall (0.8690), although precision was slightly lower (0.7807), resulting in an F1 score of 0.8225. They demonstrate that multimodal fusion, especially with gating mechanisms, effectively integrates heterogeneous campaign information to improve predictive performance, although none of them were able to surpass the best-performing LR model that combines TF-IDF text features with metadata.

Table 10. Performance of multimodal deep learning models

	ROC-AUC	Accuracy	Precision	Recall	F1
1. Tri-Modal Residual Multi-Branch	0.8956	0.8091	0.8063	0.8236	0.8149
2. Tri-Modal Gated Fusion	**0.9011**	**0.8154**	**0.8168**	0.8226	0.8197
3. Feature-Level Gated Fusion	0.8950	0.8086	0.7807	**0.8690**	**0.8225**

In addition to traditional and deep learning approaches, a large language model (Gemini 2.5 Flash) was evaluated using a few-shot multimodal prompting strategy. The given prompt included structured campaign information, textual descriptions, and campaign title images, as well as two labeled examples (successful and failed campaigns). For each targeted campaign, the LLM generated a binary prediction (1=success/0=failure) along with a short qualitative explanation. The model achieved an accuracy of 75.50% on the evaluated subset of campaigns. The detailed classification performance is summarized in Table 11.

Table 11. Classification report for the LLM

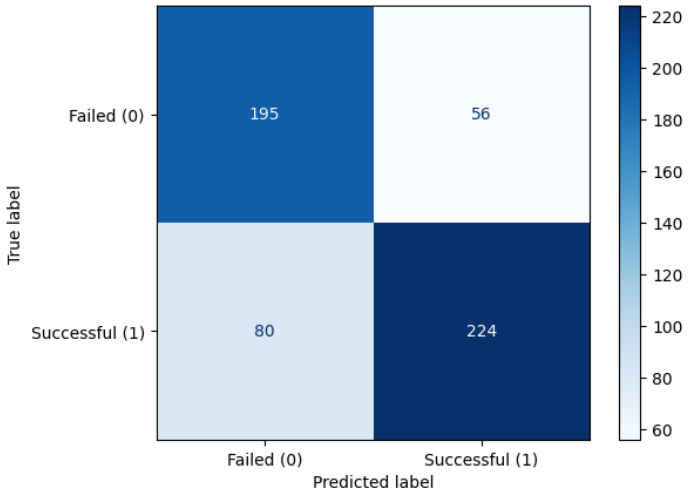
```
Classification Report:
                precision    recall  f1-score   support

    Failed (0)       0.71      0.78      0.74       251
Successful (1)       0.80      0.74      0.77       304

      accuracy                           0.75       555
     macro avg       0.75      0.76      0.75       555
  weighted avg       0.76      0.75      0.76       555
```

These results show a relatively balanced performance in both classes, with a slight tendency to be better at identifying failed campaigns (higher recall rate in class 0) while being more accurate in predicting successful ones. In total, the model produced 80 false negatives (successful campaigns predicted as failures) and 56 false positives (unsuccessful campaigns predicted as successes), as seen on Figure 14. This suggests that the LLM is somewhat conservative in predicting success, missing a significant number of actually successful campaigns while occasionally overestimating weaker campaigns. However, the relatively similar macro-averaged F1 score (0.75) confirms that the performance is not heavily biased toward one class.

Figure 14. Confusion matrix for the LLM predictions



For a systematic analysis of the qualitative explanations generated by the LLM, BERTopic was used to extract semantically coherent topics from the textual outputs. In contrast to traditional topic modeling approaches, BERTopic uses transformer-based embeddings combined with clustering techniques to identify topics based on semantic similarity (Grootendorst, 2022). This makes it particularly suitable for short texts such as explanations generated by LLMs. Two main topics emerged, such as Topic 0: Campaign Failures and Topic 1: Campaign Successes. The word cloud (Figure 15) suggests that the model often attributes potential failure to a lack of visual content, high funding goal, and overall poor campaign presentation. Representative explanations often cite “critical lack of visual assets”, “extremely ambitious goal“, “low-quality images”, and “insufficient reward structure” as reasons for failure. This supports previous work that highlights the importance of visual and reward features in attracting supporters. On the other hand, the LLM generally associates success with well-defined funding goals, credible and experienced creators, and the presence of tangible rewards (Figure 15). Explanations in this topic often mention the phrases “highly likely to succeed due to a clear funding goal” or “creator’s proven track record”, reflecting the model emphasizes goal achievability, creator credibility, and campaign quality in its predictions. The identified topics are in close alignment with the key features derived from the quantitative models, especially the funding goal and campaign presentation variables. Moreover, they are perfectly in line with prior research on crowdfunding success factors (Mollick, 2014; Kunz et al., 2017; Song et al., 2019; Baber, 2020).

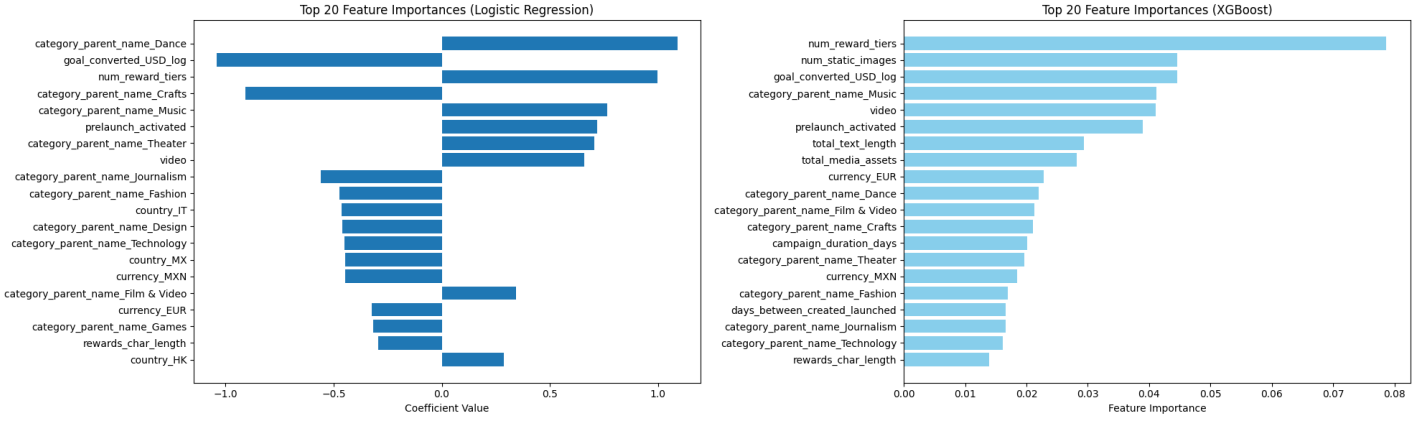
Figure 15. Word clouds of Topic 0 (Failed Campaigns) and Topic 1 (Successful Campaigns)



Feature Importance

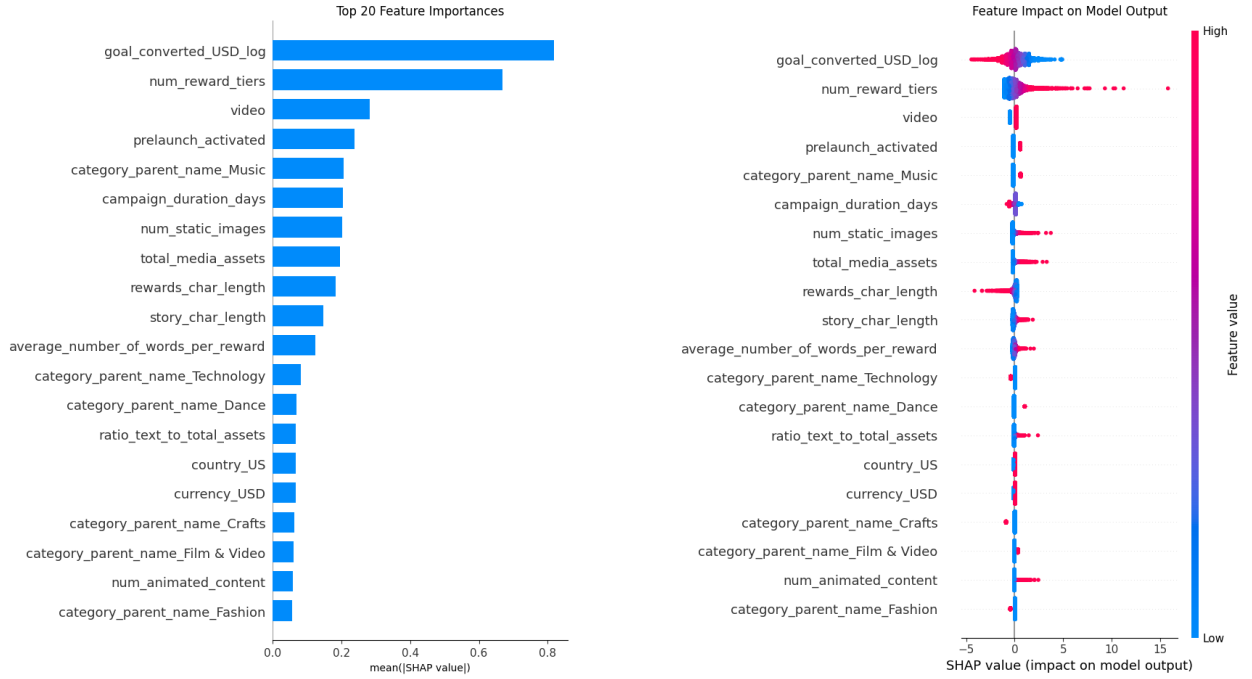
For the best-performing models, feature importance was examined in two ways, directly from model coefficients and using SHAP values. In case of Logistic Regression, the top features by model coefficients indicated that certain categorical and campaign-specific features were strongly associated with the target outcome (Figure 16). The results were clearly dominated by categorical features representing country and campaign categories, such as Dance and Music, as well as some other structural campaign elements like number of reward tiers and goal amount. It is important to note that these results are based solely on metadata features and metadata-only models. However, this clear dominance of categorical features is partly a result of how one-hot encoding assigns the effect of each category to a separate variable, which can inflate the magnitude of the coefficients. For XGBoost the top features based on the model's native importance metric were dominated by structural and campaign-specific elements, including the number of rewards tiers, the number of static images, the campaign goal amount, the presence of a video, and pre-launch activation. Other features such as total text length, total media assets, and categorical indicators of campaign categories and currency were also among the most influential (Figure 16).

Figure 16. Most important features of LR and XGBoost (metadata only) by coefficients



In contrast, SHAP values for LR highlighted characteristics that were more directly related to campaign success. The most important features were the funding goal (higher funding goals typically reduced the likelihood of success), the number of reward tiers, the presence of a campaign video, and pre-launch activation, all of which increased the likelihood of success (Figure 17). SHAP also highlighted the importance of content-related features such as the campaign duration, story length, and number of images or media assets, which were less prominent in the raw coefficients but contributed systematically to individual predictions. By taking into account interactions between features and the distribution of values across campaigns, SHAP provided a detailed view of how different aspects of campaign design and delivery influence results, making it particularly valuable in understanding the drivers of action that drive success.

Figure 17. Most important features of Logistic Regression (metadata only) by SHAP values

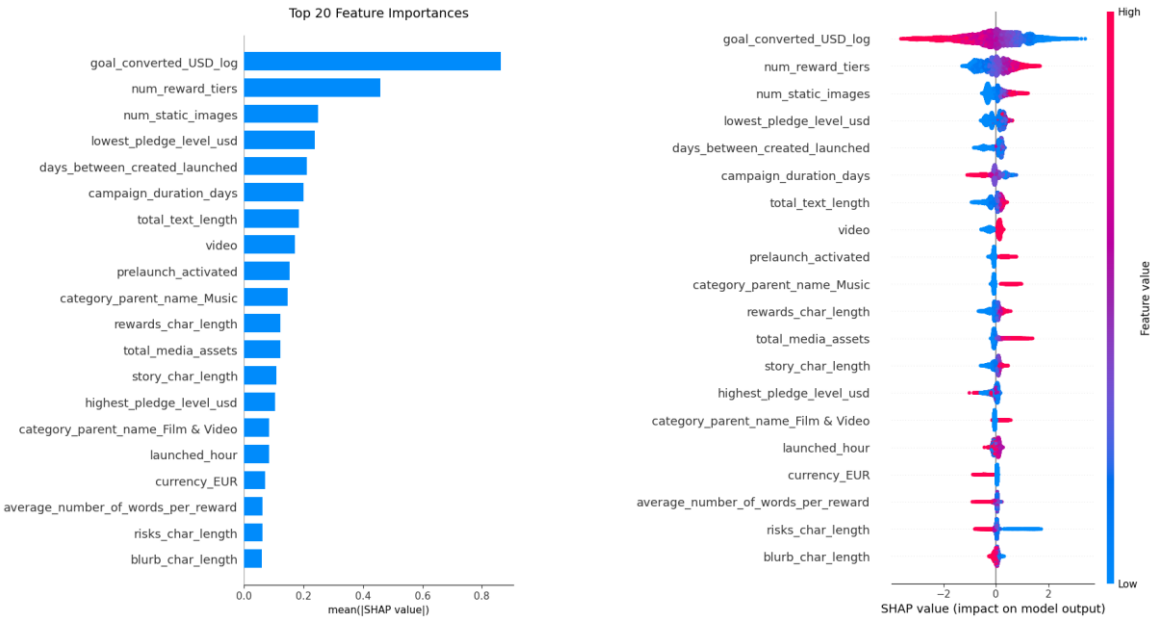


For XGBoost the built-in importance scores highlight which features were used most frequently by the trees and how much they contributed to splitting decisions, but they do not indicate the direction of influence (positive or negative) on campaign success. In contrast, SHAP values quantify both the magnitude and direction of each feature's impact on individual predictions. Using SHAP, the goal amount was found to be the most influential feature again, with larger funding goals generally decreasing the likelihood of success (Figure 18). Features that increased the likelihood of success included the number of reward levels, pre-launch activation, the presence of a campaign video, and structural content characteristics such as campaign duration, number of days between creation and

launch, total text length, and number of static images or media content. Categorical campaign indicators such as Music and Film & Video were also included. SHAP also identified features related to rewards content (e.g., reward description length) as consistently contributing across campaigns.

The emphasis on structural campaign features is quite in line with prior work on factors of crowdfunding success for both Logistic Regression and XGBoost. In particular, the large and negative consistent effect of larger funding goals supports the findings of Mollick (2014), Baber (2020) and Lin & Boh (2021), who find that smaller goals reduce the perceived risk and increase the likelihood of success. The positive effect of the number of reward levels is also consistent with previous studies (e.g., Mollick, 2014; Shneor et al., 2021; Li et al., 2022; Lin & Boh, 2021) as well as the concept of an optimal range suggested by Elitzur et al. (2024). Media-related characteristics, such as the presence of a campaign video and the number of images, further reinforce previous evidence that high-quality visual content increases credibility and helps reduce supporters' uncertainty (Mollick, 2014; Koch & Siering, 2015; Abuamer et al., 2025). Similarly, the relevance of the campaign duration identified in the SHAP analysis supports the findings of Song et al. (2019) and Elitzur et al. (2024), which show that shorter, more focused campaigns generally perform better.

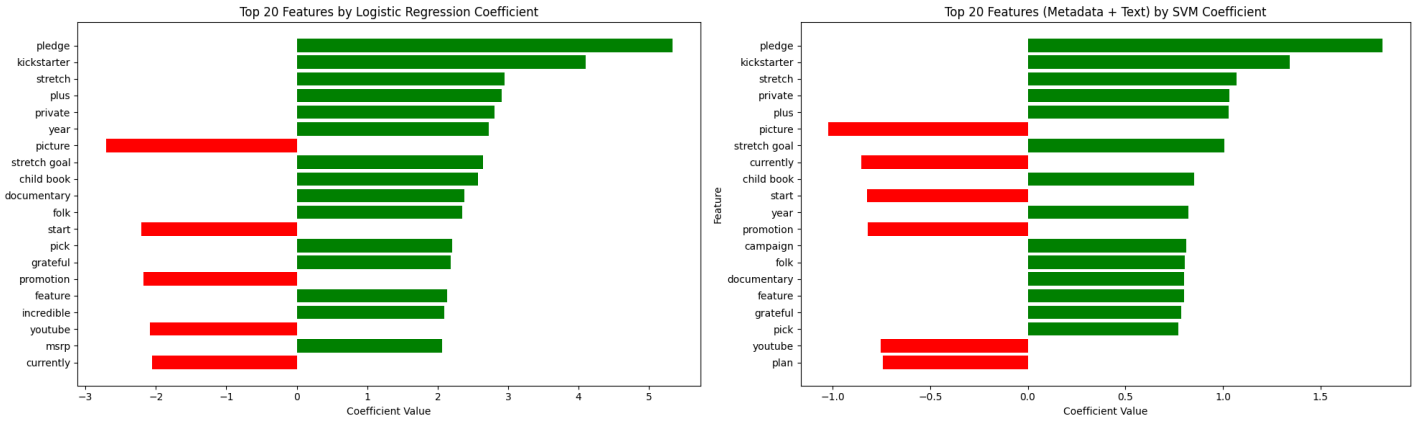
Figure 18. Most important features of XGBoost (metadata only) by SHAP values



Feature importance was further analyzed for the metadata + text models as well, particularly focusing on the best-performing ones, such as the Logistic Regression and SVM models combining TF-IDF text representations with structured metadata. For Logistic Regression, the most important features in terms of coefficient magnitude were predominantly textual, reflecting key campaign terms such as

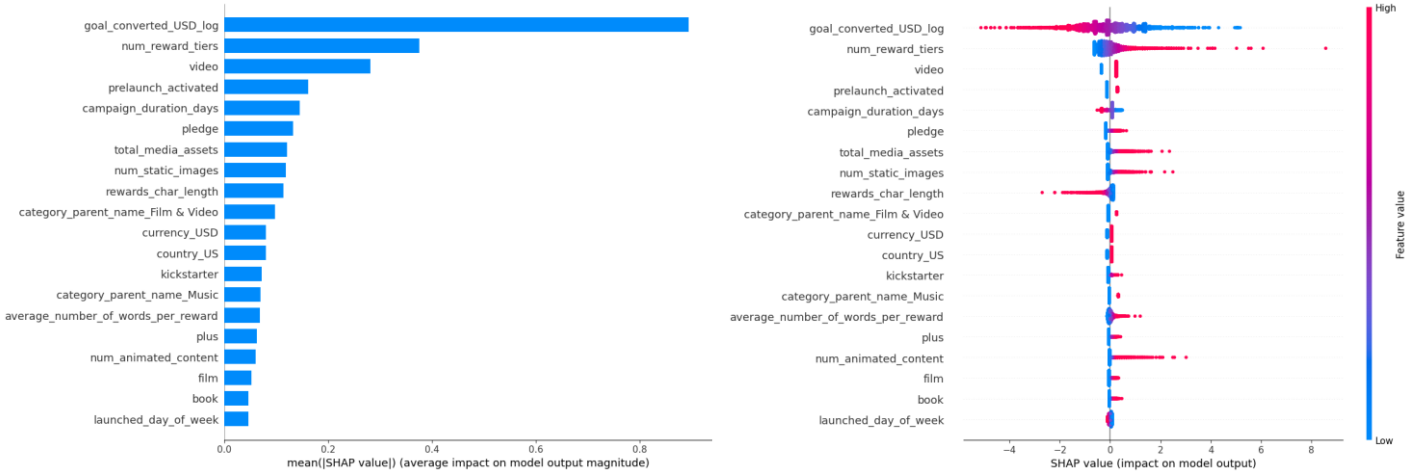
"*pledge*", "*kickstarter*", "*stretch*", "*stretch goal*" and "*plus*", as well as some negative indicators such as "*picture*", "*start*" , and "*youtube*" (Figure 19). A similar pattern was observed for the SVM model. The top 20 features in terms of absolute coefficient magnitude again emphasized text tokens such as "*pledge*", "*Kickstarter*", "*stretch*", and "*plus*", while "*picture*", and "*promotion*" contributed negatively. As these token-level effects are somewhat context-dependent, their interpretation will be discussed in more detail in the Discussion section.

Figure 19. Most important features of Logistic Regression and SVM (TF-IDF + metadata)



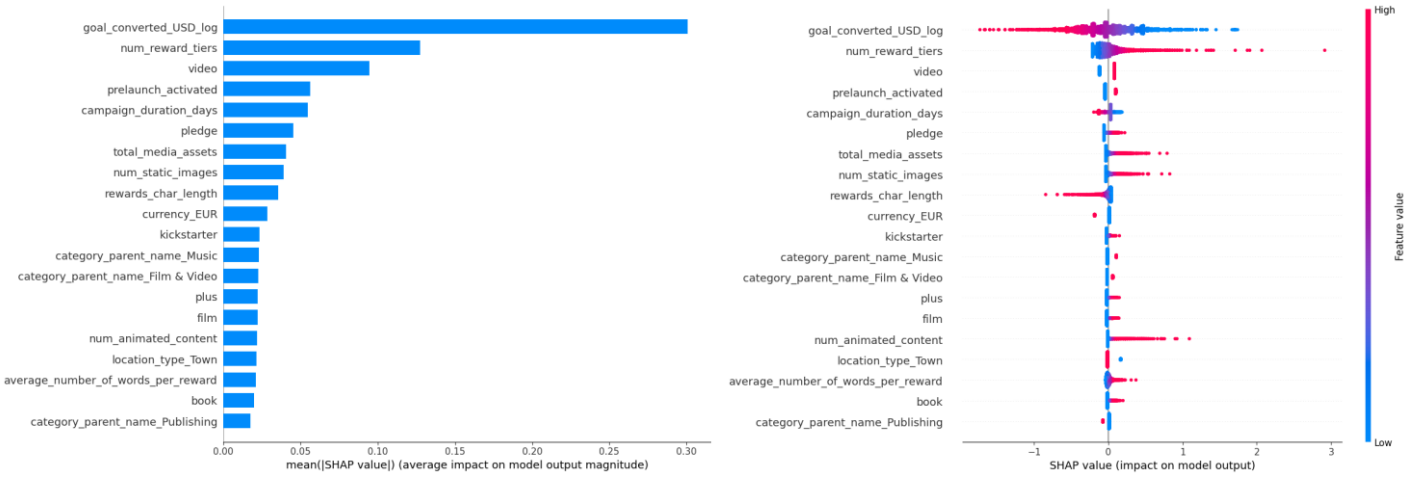
When considering average absolute SHAP values, metadata attributes such as *goal_converted_USD_log*, *num_reward_tiers*, and *video* emerged as the most powerful predictors once again, followed by content-related attributes such as campaign duration, media assets, and specific text tokens (pledge, Kickstarter) (Figure 20). This points to the combined effect of campaign structure and language on predicted success.

Figure 20. Most important features of Logistic Regression (TF-IDF + metadata) by SHAP values



For the SVM model, SHAP-based importance again highlighted structured metadata as the most influential factors, such as funding goal, number of reward tiers, and presence of a campaign video, with textual features contributing moderately to the outcome (Figure 21).

Figure 21. Most important features of XGBoost (TF-IDF + metadata) by SHAP values



6. Discussion

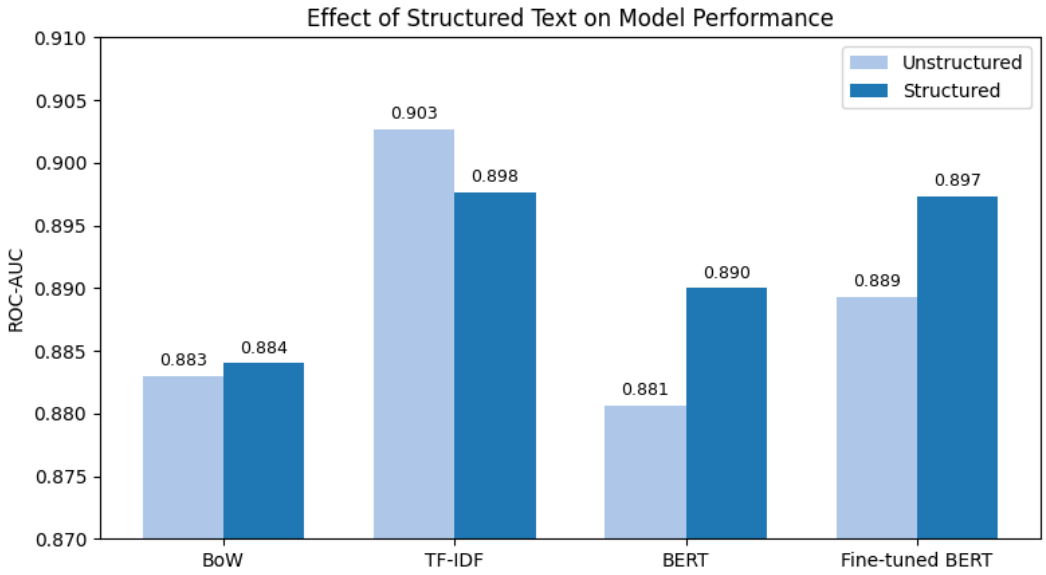
An analysis of the different input modalities reveals clear differences in predictive performance. Models that only use structured metadata already perform well, with models such as CatBoost, XGBoost, and LightGBM, achieving an ROC-AUC score of approximately 0.88. This level of performance is competitive with or exceeds many previously published pre-launch models, especially those that rely on more complex deep learning architectures. This indicates that much of the predictive information can be captured by carefully designed metadata features even without text or images.

In the evaluated models, linear methods such as Logistic Regression or SVMs always obtained comparable or even superior results to complex models. This is particularly evident in the combined text + metadata setting, where linear models using TF-IDF features achieved the highest ROC-AUC scores (up to 0.904), while delivering consistent performance on both the training and validation sets. Integrating textual information with structured data enables predictive models to evaluate campaign success more effectively than if they relied solely on metadata. Compared to previous works on multimodal success prediction on crowdfunding campaigns, this result is particularly strong, especially in light of that several recent studies using deep learning or multimodal architectures show lower or similar ROC-AUC values despite using models with significantly greater complexity. The strong performance of these models in this study can be attributed to the informative high-

dimensional feature space. Representations, like TF-IDF, convert textual data to sparse vectors, where relevant signals are well separated. This makes it suitable for efficient classification using linear decision boundaries. In such settings, the target variable appears to be mostly additively related to the features, reducing the need to model more complex nonlinear relationships. More flexible models that can model more complex interactions like Random Forest and XGBoost also showed signs of overfitting, indicating that more complexity does not always mean better generalizability in this task.

In the comparison of different text representation methods, TF-IDF always outperforms both Bag-of-Words and BERT-based embeddings. BoW representations achieve competitive, but slightly lower performance, while BERT embeddings do not provide a clear advantage and often underperform compared to TF-IDF. It shows that simple keyword-based features from TF-IDF are especially useful for this task, while more complex contextual embeddings do not seem to offer much additional value here. Additionally, two different text representation techniques were compared, an unstructured approach, where textual features were combined into a single document, and a structured approach, where each text feature was processed separately. It is interesting to see that the results do not show a consistent dominance of either approach. While unstructured representations often achieved the highest overall performance, structured representations performed competitively and in several cases outperformed their unstructured counterparts when the same underlying vectorization technique was used. This suggests that both approaches capture different aspects of the data. This behaviour is illustrated in Figure 22, which shows the average ROC-AUC scores across LR, SVM and XGBoost for structured and unstructured text representations under different encoding methods.

Figure 22. Comparison of performance between text representation methods



The unstructured representations have a richer and more compact feature space with stronger statistical signals that allows the models to capture patterns that span multiple elements of a campaign. Whereas structured representations preserve the semantic differences between different fields of text, which can be beneficial if field-specific information carries predictive value. However, the advantages of structural representations are not consistent across model and feature types. Splitting text into multiple fields increases feature sparsity and may dilute individual signals, especially for linear models with limited ability to capture interactions between feature groups. The results show a trade-off between capturing general text patterns and preserving section-specific information.

The Simple Concatenation Fusion model performed well as a baseline (ROC-AUC 0.8867, F1 score 0.8078) for neural networks using metadata and text, indicating that a simple concatenation model captures most of the predictive signal. Adding more depth helps the network better capture nonlinear relationships, as seen in the Deep Fusion MLP model, which achieved slightly higher ROC-AUC (0.8909) and F1 scores (0.8164). More advanced models with deeper interaction layers can better extract subtle correlations between metadata and textual features. The Gated Fusion Network further improves the performance by dynamically weighting the contributions of each modality. This model achieves an ROC AUC of 0.8916 and an F1 score of 0.8190 by adjusting the effects of metadata and text sample by sample. It can account for the fact that some campaigns rely more heavily on textual or structured information. Interestingly, the Cross-Interaction Network that explicitly models the interactions between individual metadata and textual features using a diamond-shaped multi-layer perceptron achieves a similar ROC-AUC value of 0.8904 but a slightly lower recall rate (0.8233). This implies that capturing detailed interactions is useful, but it does not always lead to improvements across all performance metrics. However, multi-branch architectures that encode text segments separately always perform better than simpler and less structured models. The Multi-Branch Sectional Network obtained an ROC-AUC of 0.8993 and an F1 score of 0.8235, showing the benefit of maintaining section-specific information. The best ROC-AUC (0.9002) and accuracy (0.8144) are achieved by the Residual Multi-Branch Network with direct residual connections from the metadata branch to the late-stage layers. Moreover, the residual connections help the network preserve useful metadata information in the deep fusion, so it is crucial to keep key structured features along with the detailed text representations. These results are closely consistent with previous observations regarding structured text representations. In particular, neural networks are better equipped to handle the increased complexity and sparsity that comes from dividing text into multiple sections than linear models, as they can learn interactions between representations through shared dense layers. As a

result, the disadvantages of structured representations observed in linear models are mitigated in this environment, and the models are able to utilize the additional granularity that leads to better performance. However, improvements remain relatively modest, consistent with the previous conclusion. While multi-branch architectures demonstrate the value of preserving section-specific information, the strong performance of simpler fusion models suggests that global text representations already capture a large part of the predictive signal.

It is also important to note that none of the neural network architectures are able to outperform the best linear model. The Logistic Regression model using TF-IDF features and metadata achieved the highest overall performance, with a ROC-AUC of 0.904 and an F1 score of approximately 0.82, outperforming all deep learning approaches. The Residual Multi-Branch network achieves a similar performance (ROC-AUC 0,9002) but it does not exceed it. This is in line with the findings of Kaminski & Hopp (2020), who also found that LR performed the best. This further strengthens that TF-IDF features are highly informative and well-separated and linear models are able to capture feature-target relationships effectively without complex nonlinear transformations in such a high-dimensional feature space. Moreover, neural networks provide only limited additional performance gains, even though they can model interactions and hierarchical representations. This once again highlights the trade-off between the model's complexity and its performance. Although the neural networks in this study use advanced fusion strategies like gating mechanisms, cross-interactions, and multi-branch architectures, their performance gains over simpler methods are limited, but they require significantly more computational resources and fine-tuning. The strong performance of Logistic Regression thus indicates that, with high-quality feature engineering, such as the combination of metadata and TF-IDF, simpler models can be more efficient and effective.

The results of multimodal deep learning networks highlight the potential of combining heterogeneous data sources to predict crowdfunding outcomes. All three examined architectures performed well and achieved ROC-AUC scores close to 0.90. It confirms that the integration of multiple modalities ensures a rich representation of campaign characteristics. Particularly, the Tri-Modal Gated Fusion Network performed best, highlighting the benefit of adaptive weighting mechanisms that enable the model to prioritize the most informative modality for each campaign. Similarly, the Feature-Level Gated Fusion model achieved the highest recall rate (0.8690), suggesting that fine-grained feature interactions may improve the model's ability to identify successful campaigns, albeit at the cost of slightly lower accuracy. The results highlighted here exceed those reported by Cheng et al. (2019) that could suggest that the proposed trimodal architectures and fusion strategies could be more

effective at capturing complementary information between modalities. However, despite these sophisticated architectures, none of the multimodal deep learning models were able to outperform the best-performing metadata + text linear baseline model either. Image representations didn't significantly improve the predictive performance, probably because textual and structured metadata already capture a large portion of the predictive signals in crowdfunding campaigns. Although images provide additional information, their contribution seems limited due to the use of generic, pre-trained embeddings, the high variability of visual quality, and the redundancy of textual descriptions. As a result, the additional complexity of incorporating image features does not result in a meaningful performance gain in this case or form.

Furthermore, the analysis of feature importances revealed an interesting difference between the Logistic Regression coefficients and the SHAP values. The coefficients were dominated by categorical variables, such as campaign category and country, mainly because these variables were one-hot encoded and not scaled. Each category or country appears as a separate binary variable, so the model attributes the effect of that group entirely to the corresponding variable (scikit-learn, n.d.). Since all categories were included in the encoding without omitting a reference level, this introduced redundancy and perfect multicollinearity among the dummy variables, which may further affect the magnitude of the coefficients in linear models (Karabiber, n.d.). This could amplify the perceived importance of these variables in coefficient-based ranking even if they have a less direct impact on the campaign's success. In contrast, the SHAP values highlighted features closely connected to campaign strategy and planning, such as the funding goal, the number of reward levels, the presence of a campaign video and the duration of the campaign, that directly influence the likelihood of success. These findings are consistent with previous research, which has consistently identified these factors as key determinants of crowdfunding outcomes (Mollick, 2014; Koch & Siering, 2015; Lin & Boh, 2021; Shneor et al., 2021; Li et al., 2022; Elitzur et al., 2024; Abuamer et al., 2025). In models combining metadata and text features, the coefficients were dominated by textual elements. This happens because text features are extremely multidimensional, and each word appears as a unique feature (Hastie et al., 2009). Even if each word has a small effect, the model coefficients may appear to be dominated by words just because of the large number of text features. Words such as "*pledge*," "*Kickstarter*," or "*stretch goal*" do not necessarily represent independent success factors but rather function as proxies for broader communication patterns. For example, phrases such as "*pledge*" and "*stretch goal*" are often used in campaigns that actively engage backers and communicate the funding process. Stretch goal, particularly refers to promotional rewards or bonus content that is often offered

after the initial goal has been reached. This shows that successful campaigns tend to use additional incentives and offer extended value to backers that can increase engagement and encourage larger contributions (Thürridl & Kamleitner, 2016). They also use language that is more action-oriented and community-focused and thereby confirm earlier studies that emphasize the importance of clear and persuasive communication (Shneor et al., 2021). Words like "*grateful*" or "*incredible*" can further convey a more positive and appealing tone, which can help build trust. Other positive phrases like "*documentary*", "*folk music*", and "*children's book*" likely capture effects specific to each category, reflecting that campaigns in certain categories consistently perform better (Abuamer et al., 2025). In contrast, negative weighted terms such as "*picture*", "*YouTube*" or "*promotion*" may refer to more general or weaker communication strategies. These results are also influenced by dataset-specific word frequencies. The model assigns large coefficients to words that best separate the two classes, even if not directly related to success. Hence, interpretations based on coefficients of textual features should be taken with caution, especially compared to SHAP-based analyses, which better represent the combined and contextual effects of textual and structural features of campaigns. SHAP values emphasized the importance of structured metadata and key elements of campaign design, which indicates that, although textual content is important, metadata consistently influences predictions and more accurately reflects the key drivers of a campaign's success. Additionally, the integration of a large language model provided further qualitative insights. It was not able to outperform the best traditional machine learning models, but it offers a clear advantage in terms of interpretability by generating human-readable explanations for every prediction. These explanations were often in line with the results of the SHAP analysis, particularly emphasizing the importance of features such as the funding goal, reward structure and inclusion of media assets once again.

7. Conclusion

7.1. Summary

This study was designed to investigate how heterogeneous multimodal data and machine learning techniques can be used to predict the success of crowdfunding campaigns, using pre-launch information exclusively. The results provide clear answers to the research questions and highlight the relative importance of different data sources, modeling approaches, and interpretability techniques.
First of all, regarding the predictive power of structured metadata, the results show that metadata alone provides a strong signal of campaign success. Models such as Logistic Regression, SVM and XGBoost achieved high predictive performance, confirming that structured campaign attributes such

as funding goal, duration, and reward structure are highly informative even in the absence of textual or visual information. Additionally, incorporating textual features significantly improves predictive performance. In particular, models combining metadata with text representations consistently outperformed models using only metadata. Among all approaches, the Logistic Regression model using TF-IDF features achieved the best overall performance. This shows that a large portion of the predictive signals in crowdfunding data can be well captured in a high-dimensional but linearly separable feature space. Furthermore, the analysis of multimodal approaches shows that the inclusion of visual features provides only limited added value. While the proposed multimodal architectures successfully integrated heterogeneous data sources and achieved strong results (ROC-AUC ≈ 0.90), none of these models outperform the best-performing text + metadata model. This suggests that much of the predictive signal is already captured from structured and textual data and that visual information is only marginally informative for this setting. It indicates possible limitations of current image representations in this context.

Analysis of feature importance shows that factors related to campaign strategy and structure primarily drive the likelihood of campaign success. In accordance with previous research, the main predictors are funding goal, number of reward tiers, campaign duration and inclusion of media materials. However, the results show that model-coefficient-based interpretation can be misleading, especially in high-dimensional or encoded feature space. On the other hand, SHAP values provide a more trustworthy and detailed perspective of each feature's contribution. This highlights the necessity of multiple interpretability methods to provide useful information. This suggests that predictive modeling coupled with interpretability tools provides deeper and more transparent insights, overcoming a critical limitation of prior work. Adding a large language model also brings an important qualitative dimension to the analysis. While they do not outperform traditional machine learning models in terms of predictive accuracy, they do offer a clear advantage in terms of interpretability. It generated consistent and interpretable explanations that were largely consistent with the quantitative results. This suggests that LLMs can play a complementary role in predictive systems by improving the transparency and accessibility of model outputs.

The findings reveal that accurate prediction of crowdfunding success is driven primarily by the quality and organization of features, rather than the complexity of the model or the integration of various data types. Well-crafted linear models, when paired with robust feature engineering, consistently outperform advanced deep learning methods, as they are able to extract the essential predictive patterns from a high-dimensional yet linearly separable feature space. Although deep

learning architectures, including the different fusion strategies and multimodal networks, were able to achieve competitive results, their improvements compared to simpler approaches remained modest. More advanced designs, such as gated or multi-branch networks, have also shown that structured representations can provide additional, fine-grained information, which neural networks are capable of leveraging. The benefits, though, are very dependent on the ability of the model to handle the increased complexity in an efficient manner and thus require careful optimization and significantly more computational effort. The results show that although multimodal modeling is able to effectively combine heterogeneous data types such as text, metadata and visual data, this does not always result in better predictive performance. Visual features provided only marginal predictive value, likely because textual and structured metadata already capture much of the relevant signal, and the use of image representations was limited. In general, these results suggest that strong feature representation is more important than model complexity and that simpler, well-tuned models can yield robust, efficient and interpretable predictions. However, combining predictive modeling with explainability techniques, such as SHAP scores and LLM-based qualitative reasoning, provides richer and more human-readable insights, addressing an important limitation of previous research.

7.2. Theoretical and Practical Contributions

This study has important theoretical and practical contributions to academic research and practice. It extends previous work by combining structured metadata, textual features and visual information into a unified modeling framework. This paper discusses various multimodal fusion techniques for classical and deep learning machine learning models. These techniques range from simple early fusion and feature concatenation to more complex designs such as gated, cross-interaction, and multi-branch architectures. It allows a complete assessment of the integration of different data types at different stages of the modeling process. Despite the marginal effect of images on the predictive performance, this paper demonstrates that multimodal integration is feasible and allows a comprehensive evaluation of campaign features.

The study implies that early funding dynamics are not necessary for an accurate prediction, pre-launch information is sufficient. This contributes to a deeper theoretical understanding of which campaign characteristics are predictive of success and highlights the separability of early-stage features for modeling purposes. This is very important since post-launch features, like the number of backers or the pledged amount, are inherently highly predictive. Therefore, excluding these features makes this prediction task much more challenging and thereby increases the methodical rigor and practical value

of the study. This research provides a comprehensive comparison between traditional machine learning, deep learning, large language models and multimodal approaches across different feature configurations. The results emphasize the critical importance of feature representation over model complexity. The combination of TF-IDF for textual data and structured metadata proved to be sufficient for achieving strong predictive performance. It challenges the conventional view that deep learning architectures always outperform simple models in high-dimensional feature spaces. One novel contribution of the study is the comparison of unstructured and structured approaches to textual data that has not been studied before. The analysis shows that unstructured representations are effective in capturing general trends across all textual data, while structured representations preserve the distinct information of each section that neural networks can exploit more efficiently. This also implies that different representations of text can provide further insights and inform practical modeling decisions.

Another important contribution of this study is the prediction of crowdfunding success using large language models. Although LLMs have been widely used for language generation or language understanding tasks, their application for predictive modeling is still under-investigated, especially for structured and multimodal inputs. This research shows that LLMs can be effectively adapted to perform prediction tasks similar to classification while also generating qualitative reasoning simultaneously to justify their outputs. They do not outperform conventional machine learning models, but they offer a significant advantage over traditional models as they can produce clear and understandable explanations. This shows the potential of LLMs as supplementary tools that could help bridge the gap between prediction and interpretation. Moreover, this study emphasizes the necessity of combining predictive modeling with interpretability techniques. The findings suggest that relying solely on model coefficients can be misleading, especially with high dimensional or encoded data. On the other hand, SHAP values and qualitative explanations produced by LLMs provide more informative and interpretable insights into the true reasons for the success of campaigns. This overcomes the main limitations of previous works and contributes to the theoretical discussion on explainable AI.

Campaign designers and platform developers can gain useful practical guidance from the study. It recognizes important variables affecting the success of a campaign, including funding goal, number of reward levels, duration of the campaign, and media content. It highlights key variables like funding goal, number of reward levels, duration of campaign and media content that determine the success of a campaign. This knowledge helps creators to develop campaigns that are more likely to succeed.

The results also indicate that simple linear models in combination with excellent feature engineering can outperform complex deep learning methods and provide robust, interpretable and computationally efficient solutions for predictive systems. These findings can be leveraged by platform designers to improve the design of campaign submission interfaces, recommendation engines, and automated feedback systems by focusing on the features that matter most for outcomes. Finally, the integration of explainability techniques and LLM-based qualitative reasoning shows that the predictive tools are able to provide not only accurate predictions but also usable, human-readable explanations, increasing transparency and usability for both the creators and the platform stakeholders. These contributions advance the theoretical understanding of the contribution of multimodal data to crowdfunding success prediction while also providing practical guidance for designing efficient, interpretable and effective predictive systems.

7.3. Limitations of the Research

Although this study presents a comprehensive and systematic analysis, there are several limitations that need to be acknowledged. First of all, although the study integrates multimodal data, the use of visual information is limited for multiple reasons. Only title images were used, while well-designed campaigns usually contain multiple images in their descriptions or even in the rewards section. Therefore, only using title images may not fully reflect the overall visual representation of a campaign. Furthermore, the image embeddings were not specially trained for campaign photos. They were generic and pre-trained because they were extracted using ResNet that is typically trained on ImageNet, a large dataset containing more than 14 million images. This means that the embeddings mostly capture general visual features like edges, textures, shapes or patterns. Whilst this is useful for general object recognition, these models were not fine-tuned nor tailored to campaign-specific imagery. Therefore, this approach might not fully capture the subtle quality, style, or persuasive elements of images and it may lead to underestimating the predictive value of visual content.

This study examined multiple approaches for text representation, including TF-IDF, structured and unstructured text representations, and fine-tuned BERT embeddings, however, some limitations still exist. While TF-IDF provides sparse and interpretable results, it only captures keyword-level information, and it is not able to consider deeper semantic relationships, context or the more nuanced wording of campaign descriptions. On the other hand, BERT embeddings can capture contextual and semantic information, but the fine-tuning was limited by the size of the dataset, which limits the model's capability to fully exploit rare or subtle language patterns. Moreover, BERT embeddings are

computationally expensive and they require careful hyperparameter tuning that can influence reproducibility and practical application. Examining structured and unstructured representations also introduces some compromises. Unstructured representations are able to capture global patterns in all text fields, but they can lose section-specific information, while structured representations preserve section-level nuances but increase the sparsity and complexity of features. Although neural networks are better able to utilize structured representations than linear models, the benefits are modest, which means that both approaches may fail to capture certain predictive signals.

Another important limitation we need to highlight is the exclusive focus on pre-launch features. This increases the practical applicability of the research, but at the same time, it neglects important post-campaign dynamics such as early funding patterns, social influence and backer engagement. They are known to be important predictors of campaign success. Furthermore, the analysis is based on data from a single crowdfunding platform, which could restrict the generalization of the results. The extent to which particular features predict success can also depend on different platform designs, user behaviour and campaign dynamics.

Deep learning models and multimodal architectures come with challenges such as hyperparameter tuning, model selection and computational limitations. Models were compared as fairly as possible, but additional optimization or training on a larger dataset could also improve the performance of more complex architectures. Moreover, despite the fact that several fusion strategies, including early fusion, gated, cross-interaction and multi-branch architectures, have been investigated, none of the models have integrated attention mechanisms that dynamically focus on the most relevant features within or across modalities. Attention-based approaches can improve the weighting of textual, visual or metadata features per sample, capturing finer interactions missed by static fusion strategies. As a result, existing fusion methods may fail to fully utilize the extra information within the multimodal inputs, which could constrain the attainable predictive performance of the models. Additionally, although large language models have been used to generate predictions and qualitative explanations, their outputs inherently depend on prompt design and may reflect biases or inconsistencies. In this study, the LLM was given two examples as part of a few-shot prompting strategy with minimal background information, but it was not fine-tuned for the task. This approach allowed the model to leverage its existing knowledge and reasoning abilities without requiring extensive task-specific training, making the method both effective and widely applicable. At the same time, it also limits the ability of LLMs to fully adapt to the nuances of predicting crowdfunding success. As a result, its outputs can be less accurate than those from fine-tuned models, and their reasonings can omit fine

details or show distortion, although in this case they are generally consistent with the quantitative results. This highlights the potential and current limitations of applying LLMs to structured forecasting tasks using few-shot prompting.

Finally, feature importances have been analyzed using both coefficients and SHAP values, but they have been only applied to traditional machine learning models and have not been extended to deep learning architectures. As a result, neural networks are still less interpretable compared to traditional models. They are capable of capturing complex interactions and achieving competitive performance, but their internal decision-making processes are less transparent. Overall, although this study provides a rigorous analysis of crowdfunding forecasting using multimodal and LLM-based approaches, these limitations must be taken into account when interpreting the results.

7.4. Future Research Directions

The results of this study and its limitations point to several directions for future research. One such direction is to improve visual representations by building context-specific embeddings that could improve multi-modal predictions and better capture the true value of visual features. The use of all campaign images and more advanced methods for image feature extraction, such as fine-tuned neural networks, vision transformers or attention-based models, could be improved and a deeper analysis of visual signals could be achieved.

Future work could extend the methods used in this study for fine-tuning BERT embeddings and comparing structured and unstructured text representations, LLMs, additional pre-training on additional datasets specific to crowdfunding, or more sophisticated architectures that combine contextual embeddings with interpretable features. These methods might be useful to capture more subtle linguistic patterns and possibly improve predictive results. Future researchers could also incorporate temporal and social dynamics. For practical reasons this study only includes pre-launch features, but financing trends and backer engagement, social media activities or platform-driven promotions may improve the predictive performance and offer a more comprehensive view of the campaign trajectories. Platform-independent studies also have a lot of potential. Expanding the analyses to include campaigns on several crowdfunding platforms, each with their own regulations, supporter communities, and campaign structures, would improve the generalizability of the findings and help identify platform-specific predictors of success.

Moreover, the multimodal fusion strategies can be extended. The introduction of attention-based mechanisms or adaptive weighting allows the models to attend to the most important features within and across modalities and may be able to capture interactions that static fusion methods fail to detect. Investigating these strategies in combination with linear and deep learning approaches may provide further insights into the optimal integration of heterogeneous data sources. Future work might also consider improving the interpretability of deep learning models by developing or applying feature importance methods for neural networks. This would enable a more direct comparison of feature contributions across all model types. Future work could explore greater use of large language models in combination with other methods of interpretable AI. Few-shot prompting has achieved promising results, and approaches such as model fine-tuning or carefully designed prompting can potentially further improve the reliability and overall effectiveness of the predictions. Additional work on methods to enhance interpretability would also improve the practical applicability of these models in forecasting. Accurate and reliable forecasts would enable campaign creators to make better informed decisions throughout the campaign period. These directions show promise to improve predictive results and practical utility of models to forecast crowdfunding success, while further exploring the relationship between multimodal information, model architecture and interpretability.

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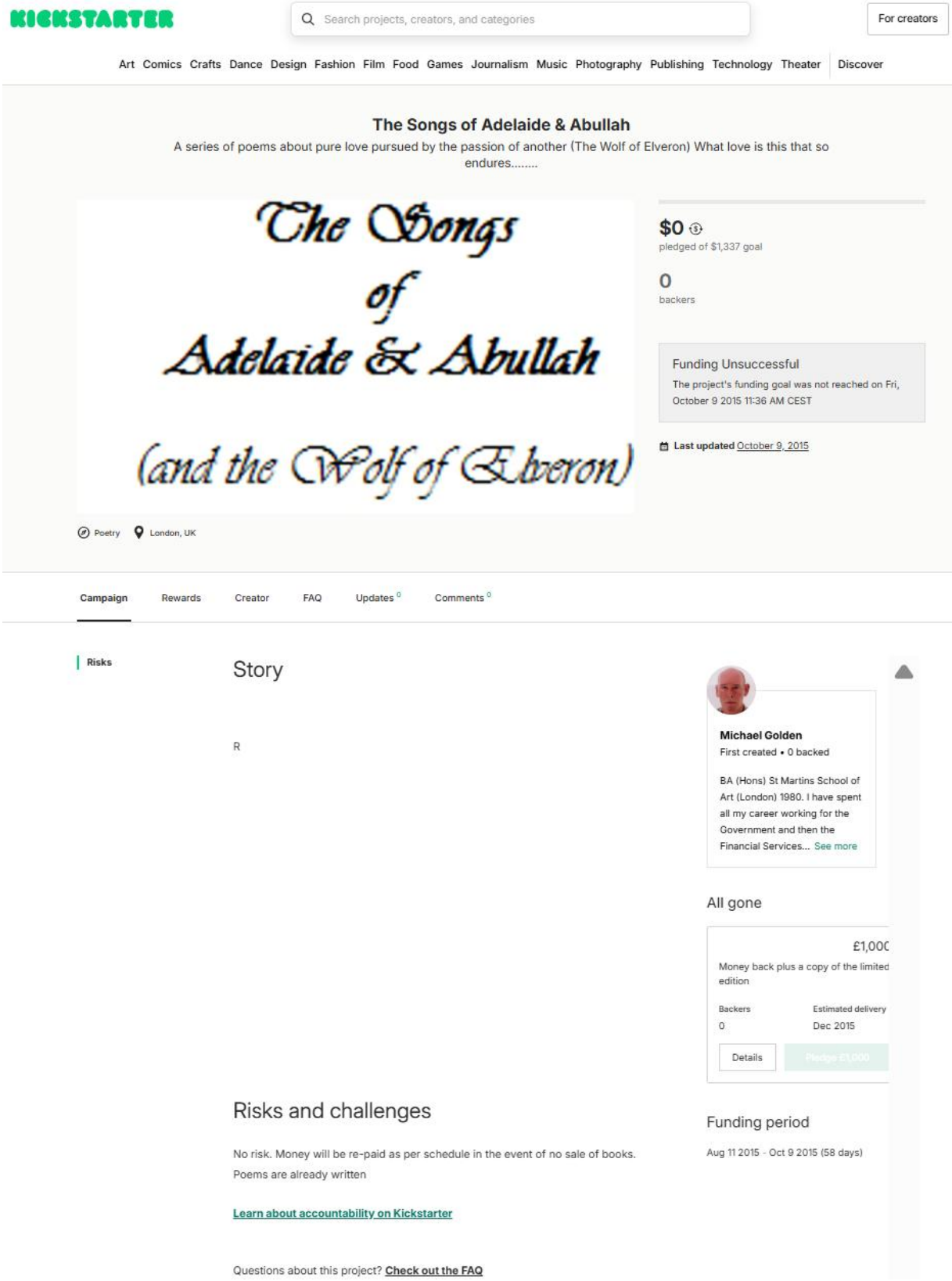
Appendix B: Abbreviation List

Abbreviation	**Full Word**
Adam	Adaptive Moment Estimation
AI	Artificial Intelligence
ANN	Artificial Neural Network
AON	All-or-Nothing
AUC	Area Under the ROC Curve
Bagging	Bootstrap aggregating
BERT	Bidirectional Encoder Representations from Transformers
BiLSTM	Bidirectional Long Short-Term Memory
BoW	Bag-of-Words
CART	Classification and Regression Trees
CNN	Convolutional Neural Network
CSV	Comma Separated Values
DCAN	Deep Cross-Attention Network
DL	Deep Learning
EDA	Exploratory Data Analysis
GloVe	Global Vectors
HTML	Hypertext Markup Language
IPT	Information processing theory
KIA	Keep-It-All
KNN	K-Nearest-Neighbour
LLM	Large language model
Log	Logarithm
LR	Logistic Regression
MDGCN	Multimodal Dynamic Graph Convolutional Network
ML	Machine Learning
MLP	Multi-Layer Perceptron
NLP	Natural language processing
NN	Neural network
OCR	Optical character recognition
RBF	Radial Basis Function
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
SHAP	Shapley Additive exPlanations
SVM	Support Vector Machine
Tanh	Hyperbolic Tangent
TF-IDF	Term Frequency Inverse Document Frequency
URL	Uniform Resource Locator
XGBoost	Extreme Gradient Boosting

Appendix C: Leakage terms that were removed from the campaign text

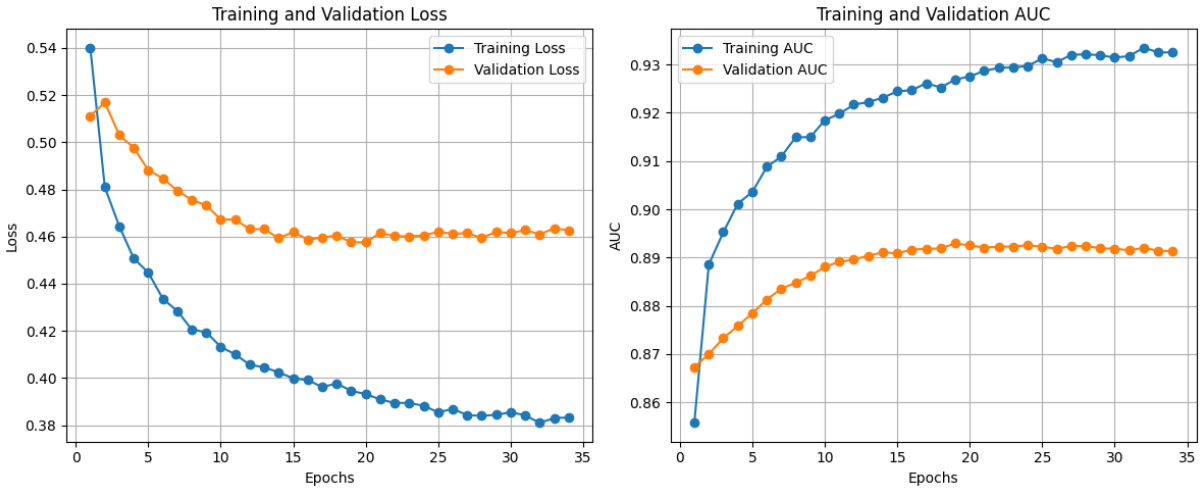
successfully funded	fully funded	campaign failed	campaign succeeded	not successful	not funded	not fully funded
campaign funded	campaign is successful	campaign is fully funded	reached our goal	goal reached	goal achieved	goal completed
reached our funding goal	funding goal reached	reached our target	target reached	target achieved	met our goal	goal met
did not reach	didn't reach	did not meet	didn't meet	we made it	we didn't make it	we did it
we didn't do it	stretch goal unlocked	stretch goal reached	stretch goal achieved	thanks for backing	thanks for pledging	thank you for backing
thank you for pledging	% funded	% of goal	% of funding goal	percent funded	percent of goal	percent of funding goal
over funded	under funded	half funded	partially funded	almost funded	nearly funded	just funded
barely funded	close to funding	close to goal	close to success	final chance	last chance	final day
last day	few days left	hours left	minutes left	seconds left	campaign ended	campaign ending
campaign closed	campaign closing	project funded	project fully funded	funding complete	funding achieved	funding successful
passed our goal	exceeded our goal	campaign completed				

Appendix D: Example of an excluded campaign

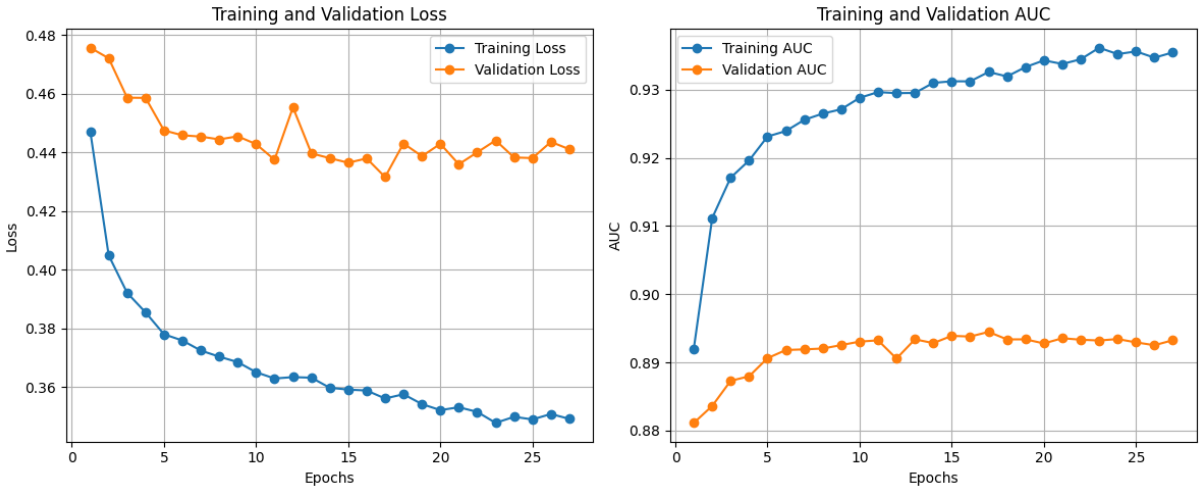


Appendix E:

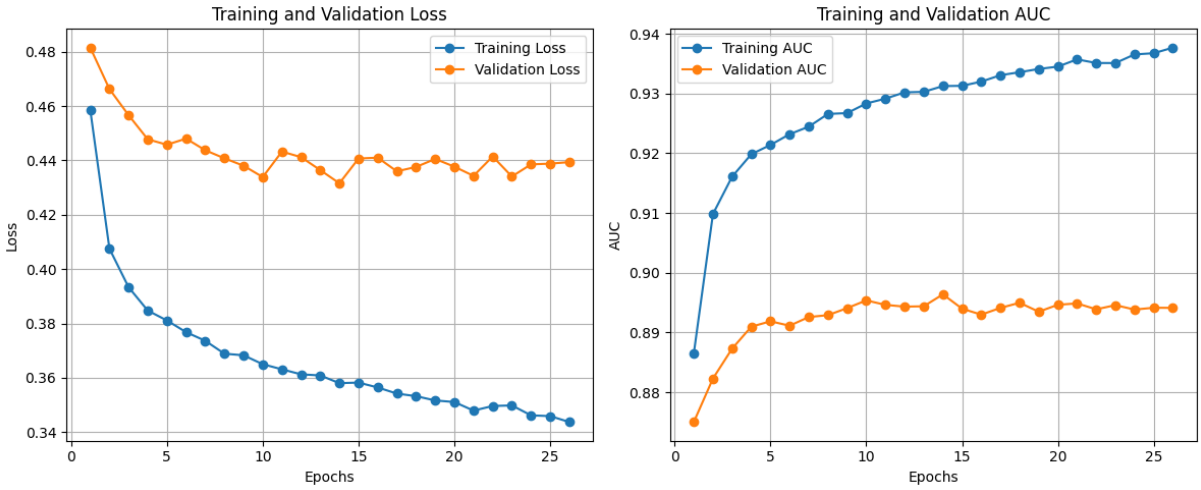
Simple Concatenation Model Training History and Performance Curves



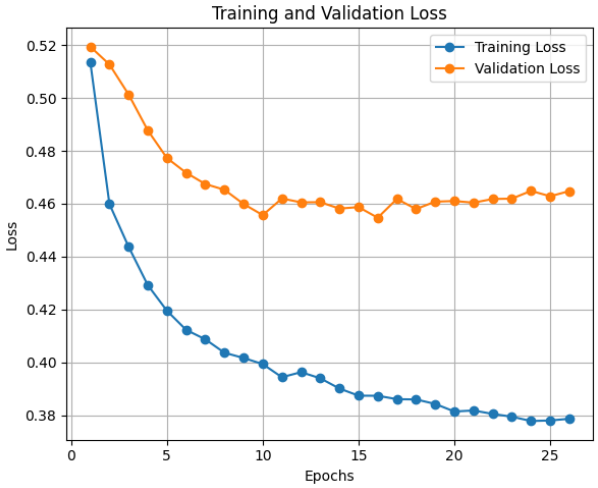
Deep MLP Fusion Model Training History and Performance Curves

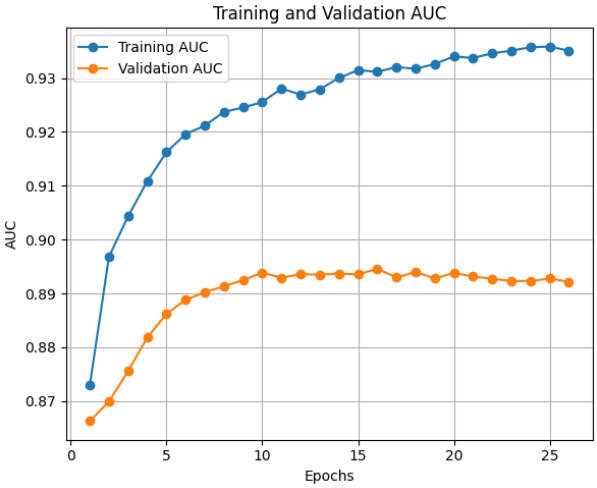


Gated Fusion Model Training History and Performance Curves

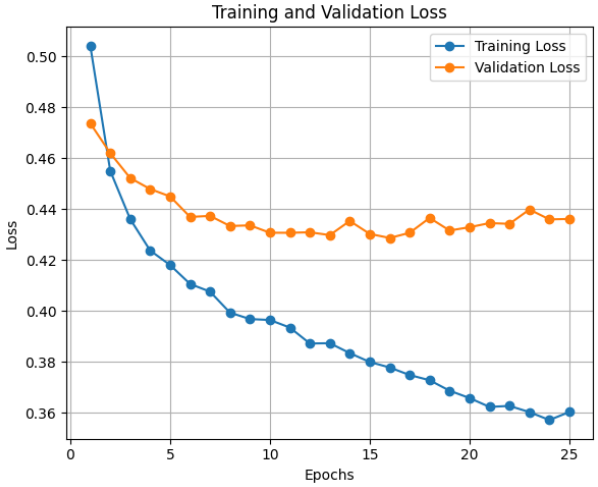


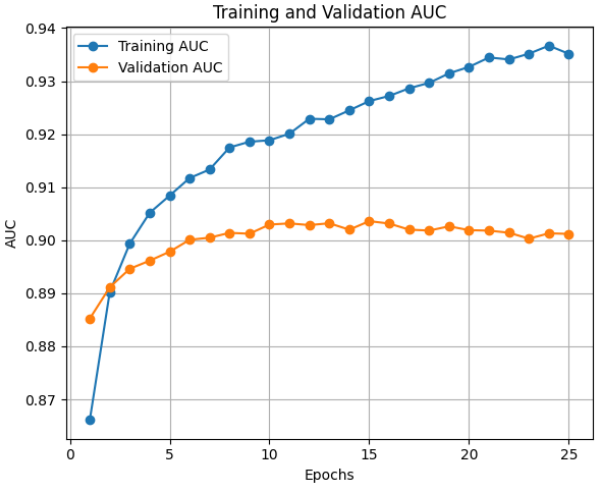
Cross-Interaction Model Training History and Performance Curves





Multi-Branch Model Training History and Performance Curves





Residual Multi-Branch Model Training History and Performance Curves

